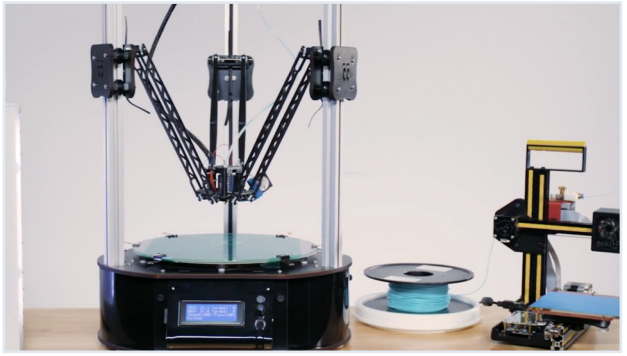


COURSE SLIDES · WSQ

Innovating Product Development with 3D Printing and Additive Manufacturing

WSQ Course Code: TGS-2022017597 · TSC: Additive Manufacturing (PRE-OPR-4012-1.1)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v8.1 · 22 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

AM · PM · Assessment

It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.

2

SSG QR code

The trainer/administrator displays the digital attendance QR code generated from the SSG portal.

3

Scan & submit

Scan the QR code with your mobile phone camera and submit your attendance each session.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer and Founder, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS
PhD — data science, engineering and emerging-technology educator.

DELIVERS
WSQ courses on 3D printing & additive manufacturing, CAD/product design, data science and AI.

EXPERIENCE
20+ years of industry training across engineering, manufacturing and technology domains.

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with CAD, 3D printing or manufacturing (if any).
- What you want to design, prototype or manufacture after this course.

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HOUSEKEEPING

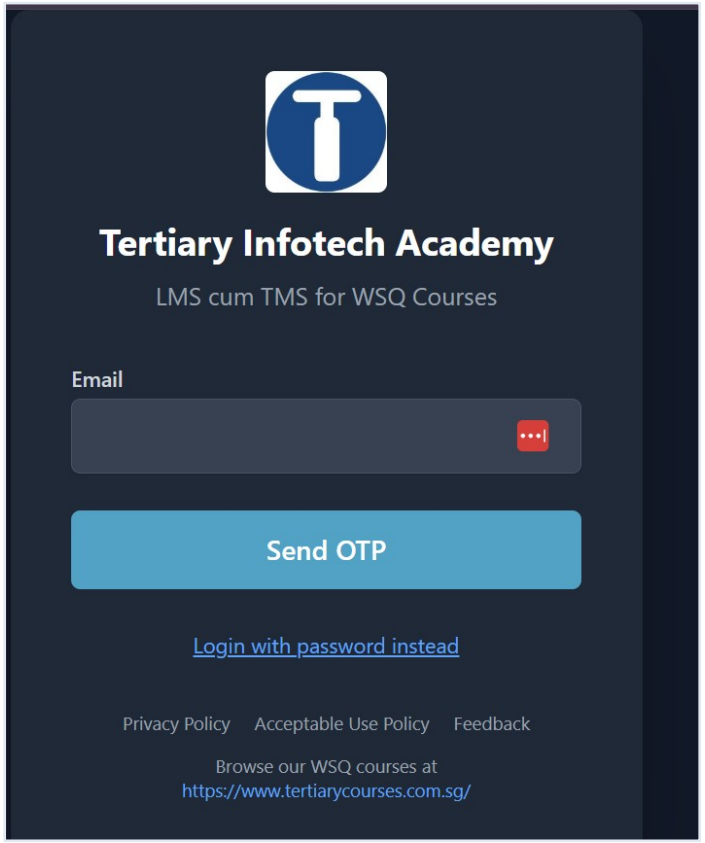
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL · LMS/TMS

Download Course Material



lms-tms.tertiaryinfotech.com — your course portal login page

1 · Visit the portal

Go to https://lms-tms.tertiaryinfotech.com in your browser.

2 · Login with OTP

Enter your registered email and key in the One-Time Password sent to you.

3 · Open your course

Select this course under My Courses.

4 · Download materials

Download the slides and Learner Guide for the open-book assessment.

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SCHEDULE

Lesson Plan — 2 Days, 8 Hours/Day

Day 1

- Day 1 — Additive Manufacturing, 3D Scanning and Design for AM
  - Digital Attendance (AM)
  - Trainer and learner introductions · Learning outcomes
  - Topic 1: Overview of AM and 3D Scanning (Labs 1–6)
  - Lunch break · Digital Attendance (PM)
  - Topic 2 begins: Process Related Issues (Lab 7)

Day 2

- Day 2 — Process Issues, Process Improvement and Assessment
  - Digital Attendance (AM)
  - Topic 2: Process Related Issues for AM (Labs 8–10)
  - Topic 3: Process Improvement for AM (Labs 11–16)
  - Revision · Course feedback · TRAQOM survey
  - Digital Attendance (Assessment) · Final Assessment (WA + PP)

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SKILLS FRAMEWORK

Skills Framework — TSC

1

TSC Title

Additive Manufacturing

2

TSC Code

PRE-OPR-4012-1.1

3

Framework

Skills Framework for Precision Engineering — Additive Manufacturing

4

Proficiency

Level 4 — analyse, determine, evaluate and optimise AM processes

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TSC Abilities

- A1: Analyse and determine material considerations of components to be manufactured
- A2: Analyse process-related issues for using DMLS, SLS, SLM, powder bed and inkjet head, and/or 3D printing for AM
- A3: Determine processes and procedures for manufacturing metallic and non-metallic components using AM
- A4: Evaluate use of reverse engineering processes for components through 3D scanning and subsequent AM
- A5: Apply optimisation techniques to improve efficiency of AM processes and product quality
- A6: Assess the impact of manufacturing process improvements on business operations

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TSC Knowledge

- K1: AM characteristics, techniques and systems
- K2: Requirements of different AM processes
- K3: Factors to be considered when selecting AM solutions
- K4: Impact of AM on manufacturing operations
- K5: Methods for reviewing AM performance
- K6: Benefits and trade-offs of AM
- K7: Methods of ensuring long-term sustainability of AM

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 — Analyse AM & 3D scanning

Analyze the AM and evaluate the use of reverse engineering using 3D scanning.

2

LO2 — Analyse process issues

Analyze the issues and determine the requirements for AM processes.

3

LO3 — Optimise & assess

Apply AM optimization techniques and assess the impact of process improvement.

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2 DAYS · 16 HOURS

Course Outline

Topic 1 · Overview

- Overview of AM systems
- Material considerations for AM
- Benefits and trade-offs of AM
- AM with 3D scanning
- Design for AM in Fusion 360
- Labs 1–6

Topic 2 · Process Issues

- Printer hardware & slicing
- Bed adhesion: skirt/brim/raft
- Extrusion & temperature issues
- Filaments and materials
- Hybrid metal + polymer parts
- Labs 7–10

Topic 3 · Improvement

- Infill and wall optimisation
- Surface quality & textures
- Support strategy
- Post-processing chains
- Business impact & sustainability
- Labs 11–16

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ASSESSMENT RULES

Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black/blue pen for hard-copy assessments.
5. No liquid paper / correction tape.
6. Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) — 1 hour, open book.

2

Practical Performance (PP)

Scenario-based practical tasks — 90 minutes, open book.

3

Open book

Slides, Learner Guide and approved materials only.

4

Eligibility

Minimum 75% attendance to be eligible for assessment and funding.

5

Appeal

An appeal process is available if required.

6

Outcome

Competent (C) / Not Yet Competent (NYC) recorded on the Assessment Summary Record.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then the Practical Performance — open book
4. Upload your answers on the LMS
5. Sign the Assessment Summary Record

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01

TOPIC 01

Overview of Additive Manufacturing and 3D Scanning

AM systems · material considerations · benefits & trade-offs · 3D scanning and reverse engineering (K1, K3, K6 · A1, A4)

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What is Additive Manufacturing?

1

Layer-by-layer fabrication

AM builds parts by depositing material layer upon layer, directly from a computer-aided design (CAD) file.

2

No part-specific tooling

No dies, molds or fixtures — the digital file is the tooling, so design changes cost nothing to retool.

3

Freeform geometry

Multiple layers built in the X–Y plane stack along Z, enabling shapes machining cannot reach.

4

Rapid & flexible

Concept-to-part in hours enables fast prototyping, iteration and mass customisation.

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TWO PHILOSOPHIES · K1

Additive vs Subtractive Manufacturing

Add material

- Additive (AM)
  - Builds the part up layer by layer
  - Material only where needed — minimal scrap
  - Complexity is essentially free
  - No tooling — economic at lot size one
  - Slower per part at high volumes

Remove material

- Subtractive (CNC/machining)
  - Cuts material away from a solid block
  - Significant chip/scrap waste
  - Complex internal features often impossible
  - Tooling & setup favour large volumes
  - Excellent finish and tight tolerances

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TERMINOLOGY

3D Printing vs Additive Manufacturing

1

3D printing

Strictly, the build step — the machine depositing layers. The term the consumer world uses.

2

Additive manufacturing

The full industrial workflow: design, simulation, build, post-processing, inspection and certification.

3

Same core, different scope

Every AM part is 3D printed — but AM adds the engineering discipline around the print.

4

In this course

We use both terms; Topic 2–3 cover the process discipline that turns printing into manufacturing.

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MILESTONES

A Brief History of 3D Printing

1. 1981 — Hideo Kodama publishes the first layer-by-layer photopolymer method
2. 1984–87 — Chuck Hull invents SLA and ships the first commercial system
3. 1988–92 — FDM patented by Scott Crump; SLS emerges for powders
4. 2009 — key FDM patents expire; RepRap & desktop printing explode
5. Today — metal AM flies in aircraft, implants and rockets

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CAD TO PART

The 3D Printing Workflow

1. DESIGN — model the part in CAD (Fusion 360) or capture it by 3D scanning
2. EXPORT — tessellate to an STL/3MF mesh file
3. SLICE — the slicer (Cura) converts the mesh to layer toolpaths (G-code)
4. PRINT — the machine builds the part layer by layer
5. POST-PROCESS — remove supports, cure, finish, inspect

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TECHNOLOGY MAP · K1

The 7 AM Process Families (ISO/ASTM 52900)

1

Material Extrusion

FDM/FFF — molten filament laid down in beads. Desktop standard.

2

Vat Photopolymerisation

SLA, DLP, LCD/MSLA — light cures liquid resin, layer by layer.

3

Powder Bed Fusion

SLS, DMLS, SLM, EBM — laser/e-beam fuses powder layers.

4

Material Jetting

PolyJet/NanoParticle jetting — droplets of material UV-cured in place.

5

Binder Jetting

Liquid binder glues powder — metals, sand moulds, full-colour parts.

6

Directed Energy Deposition

Blown powder or wire melted by laser/arc — repair & large metal parts.

7

Sheet Lamination

LOM, ultrasonic consolidation — sheets bonded and cut.

8

Hybrid systems

AM + CNC finishing in one machine — additive build, subtractive precision.

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Vat Photopolymerisation & Powder Bed Fusion

1

SLA

A UV laser cures resin one small spot at a time — fine detail, smooth finish.

2

DLP / LCD (MSLA)

A projector or LCD mask cures a whole layer at once — speed at pixel resolution.

3

SLS

A laser sinters polymer powder — tough nylon parts, no supports needed.

4

LPBF / DMLS / SLM

Laser powder bed fusion of metals — sintered (DMLS) or fully melted (SLM).

5

EBM

An electron beam melts metal powder in vacuum — aerospace titanium.

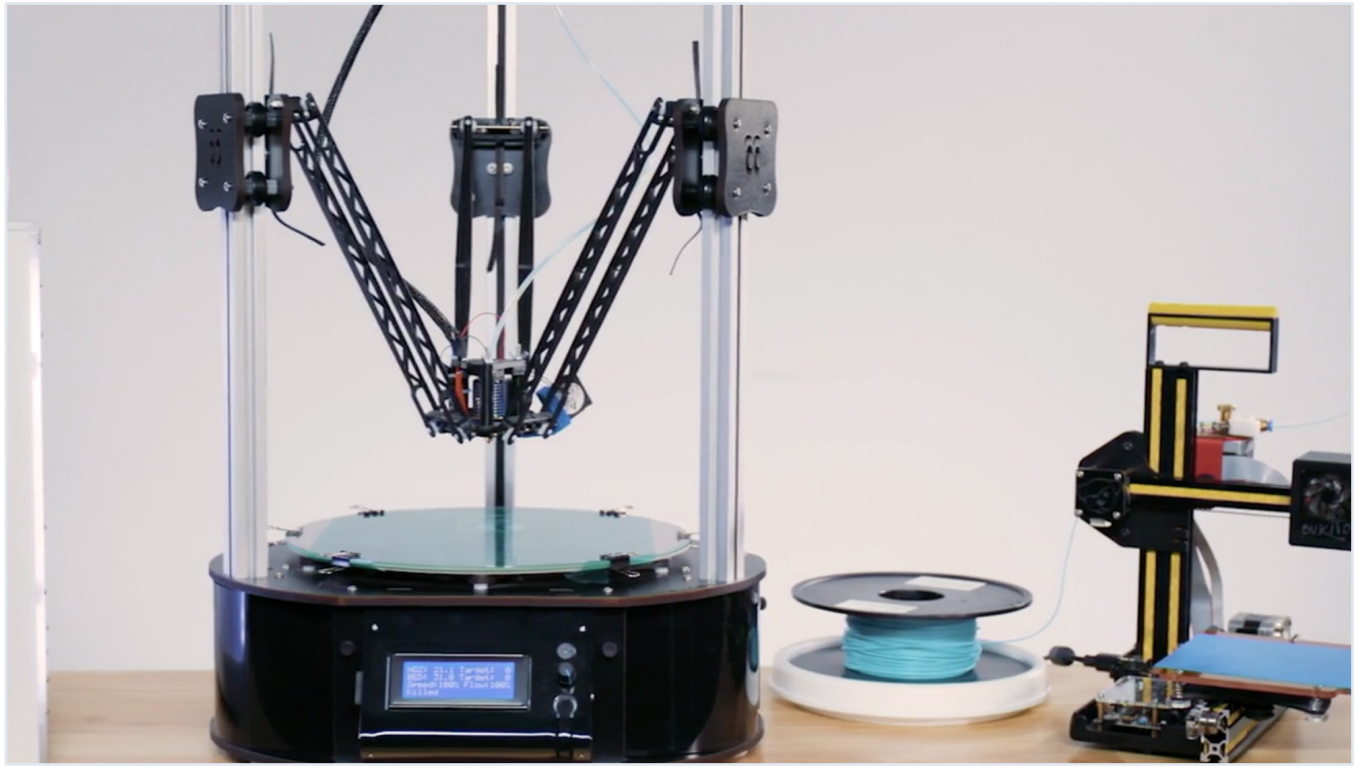
6

MJF

HP Multi Jet Fusion — agent + IR lamps fuse nylon powder fast.

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Printing with Filament (FDM)



Desktop FDM printers — cartesian and delta architectures

Layer on layer

Each layer prints on top of the previous one, up from the build platform — like building a wall.

Needs a base

The first layer must stick to the platform; poor bases need adhesion aids or redesign.

Overhangs need support

Sudden overhangs print into mid-air — the slicer generates removable support structures.

Design to avoid supports

Orient and design parts to minimise supports: faster prints, better surfaces.

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Printing with Resin (SLA / DLP / LCD)



A desktop SLA resin printer with its UV-blocking cover

UV-curable resin

A tray of liquid photopolymer is selectively hardened by a light source.

Laser vs projector

SLA cures spot by spot; DLP/LCD cure an entire layer at once — features as fine as a pixel.

Printed solid

Resin parts usually print solid or drained — trapped liquid resin cannot stay inside.

Post-cure required

Wash in IPA, then UV post-cure to reach full strength — always with gloves.

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TOPIC 01 · POWDER

Printing with Powder (SLS / Metal PBF)



An SLS part emerging from its unsintered powder cake

Most aggressive complexity

Unfused powder supports the part, freeing the most complex geometries in AM.

Industrial territory

Powder is messy and metal powders are an explosion hazard — not a casual in-house setup.

Post-processing heavy

Depowdering, heat treatment and finishing depend on technology and material.

Nylon to titanium

From SLS nylon ducting to DMLS titanium implants and rocket injectors.

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DMLS · SLS · SLM · K1/K2

Metal Powder Bed Fusion — Three Siblings

DMLS — sinter

- Laser fuses metal powder by sintering — heat without full melting
- Stainless, cobalt-chrome, aluminium, Inconel
- Dental implants, aerospace brackets

SLS — polymers

- Same principle applied to thermoplastic powders
- Tough nylon (PA12) parts
- No support structures — powder is the support

SLM — full melt

- Laser fully melts the metal to a liquid state
- Denser parts, higher residual stress
- Demands tight process control

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MATERIALS · K2/K3

AM Materials Palette

1

Polymers

PLA, ABS, PETG, Nylon, TPU filaments; photopolymer resins; PA12 powders.

2

Metals

Stainless & tool steels, titanium, aluminium, cobalt-chrome, Inconel, copper.

3

Ceramics & composites

Technical ceramics, carbon/glass-fibre filled polymers, metal-matrix blends.

4

Specialty

Wood/metal/stone-filled, conductive, magnetic, glow, flexible TPE, bio-inks.

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Selecting an AM Solution — Decision Factors

1

Material properties

Mechanical strength, temperature, chemical resistance and biocompatibility the part must deliver.

2

Design complexity

Feature detail, internal channels, overhangs — some processes resolve far finer detail than others.

3

Volume & speed

Prototyping favours AM at lot size one; beyond ~100s of units, moulding usually wins.

4

Post-processing

Supports, curing, machining, finishing — each process carries its own downstream cost.

5

Cost structure

Machine, material, labour and energy per part — plus capital outlay and maintenance.

6

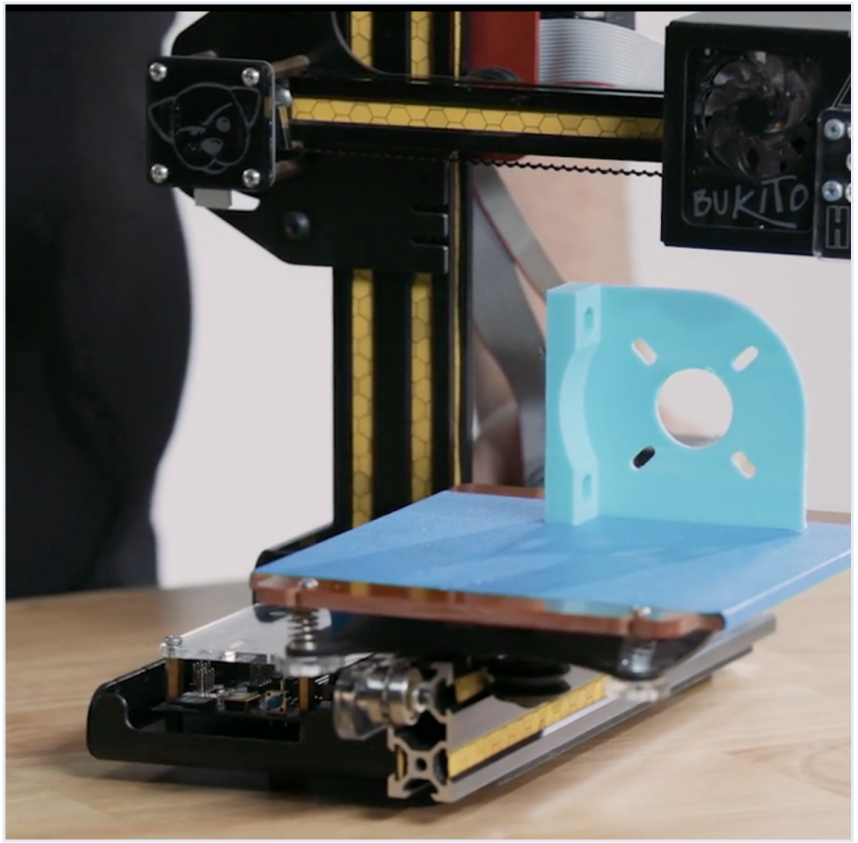
Learning curve

Operator skill, software toolchain and process maturity in your team.

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K3 · PROCUREMENT

Considerations When Buying a 3D Printer



A desktop FDM printer mid-print

Cost
Machine price, materials, spares and running cost per printed hour.

Learning curve
From plug-and-print desktop FDM to industrial PBF needing trained operators.

Build volume
Big printers cost and occupy exponentially more — match to your part sizes.

Capital & maintenance
Ventilation, power, calibration schedules and service contracts.

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Benefits of Additive Manufacturing

1

Design freedom

Complex geometry is free — internal lattices, conformal channels, organic forms.

2

Rapid prototyping

Idea to physical part in hours; iterate daily instead of quarterly.

3

Mass customisation

Every part can differ at no extra cost — dental, hearing aids, prosthetics.

4

Less waste

Material is placed only where needed — high buy-to-fly ratios beaten dramatically.

5

No tooling

Skip molds and dies entirely — economic at lot size one, instant design changes.

6

Distributed production

Ship files, not parts — print at the point of need.

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K6 · EYES OPEN

Trade-offs of Additive Manufacturing

1

Throughput

Layer-by-layer is slow — injection moulding beats AM beyond hundreds of units.

2

Mechanical anisotropy

Parts are typically 10–50% weaker across layer boundaries than moulded equivalents.

3

Material range

Far fewer qualified materials than conventional manufacturing, and often costlier.

4

Surface & accuracy

Layer lines and dimensional tolerance often demand post-processing.

5

Post-processing load

Support removal, curing, heat treatment — hidden time and labour.

6

Repeatability & QA

Process variation makes certification harder — a live research frontier (NIST).

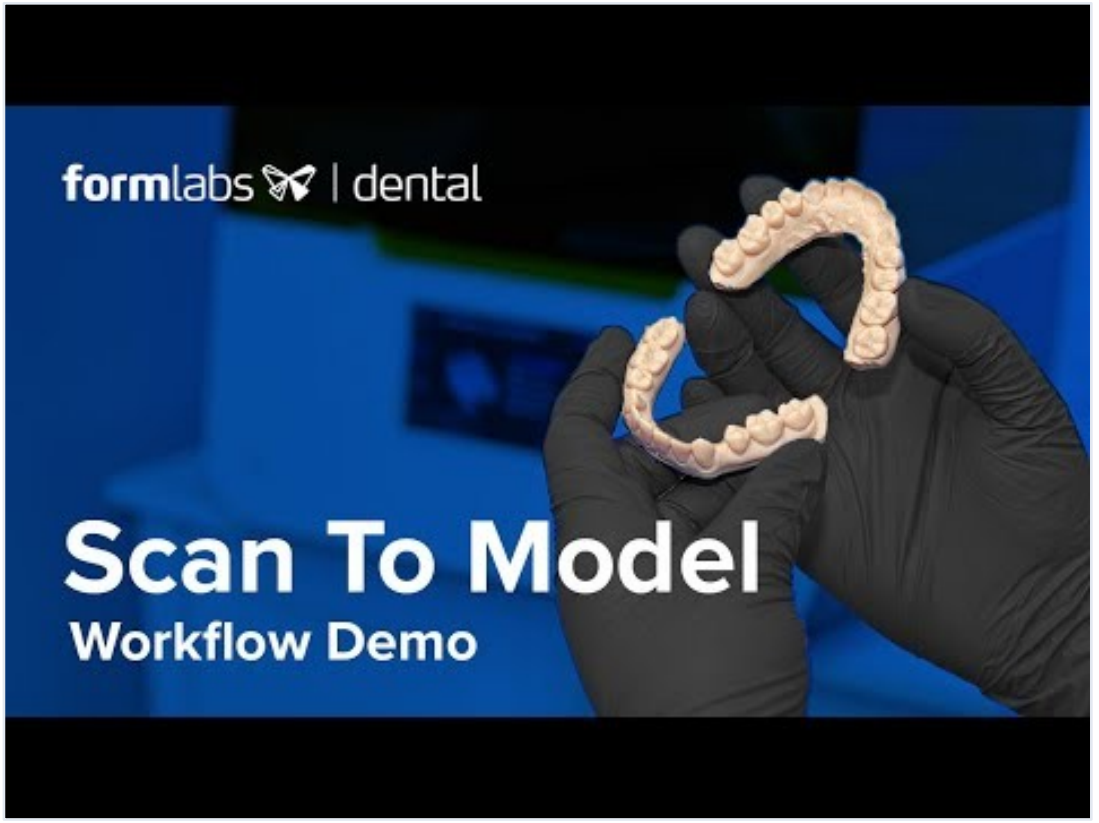
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CASE STUDY · K4/K6

Case Study — Digital Dentistry



3D-printed dental models and appliances



Scan-to-model workflow: intraoral scan → print

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CASE STUDY · MASS CUSTOMISATION · 99% OF SHELLS ARE 3D-PRINTED

Case Study — Hearing Aids



Ear impression captured for a custom fit



Custom-fit 3D-printed ear tips

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CASE STUDY · ACCESSIBLE ENGINEERING

Case Study — Prosthetics for Humans and Animals



Printed prosthetic hand components

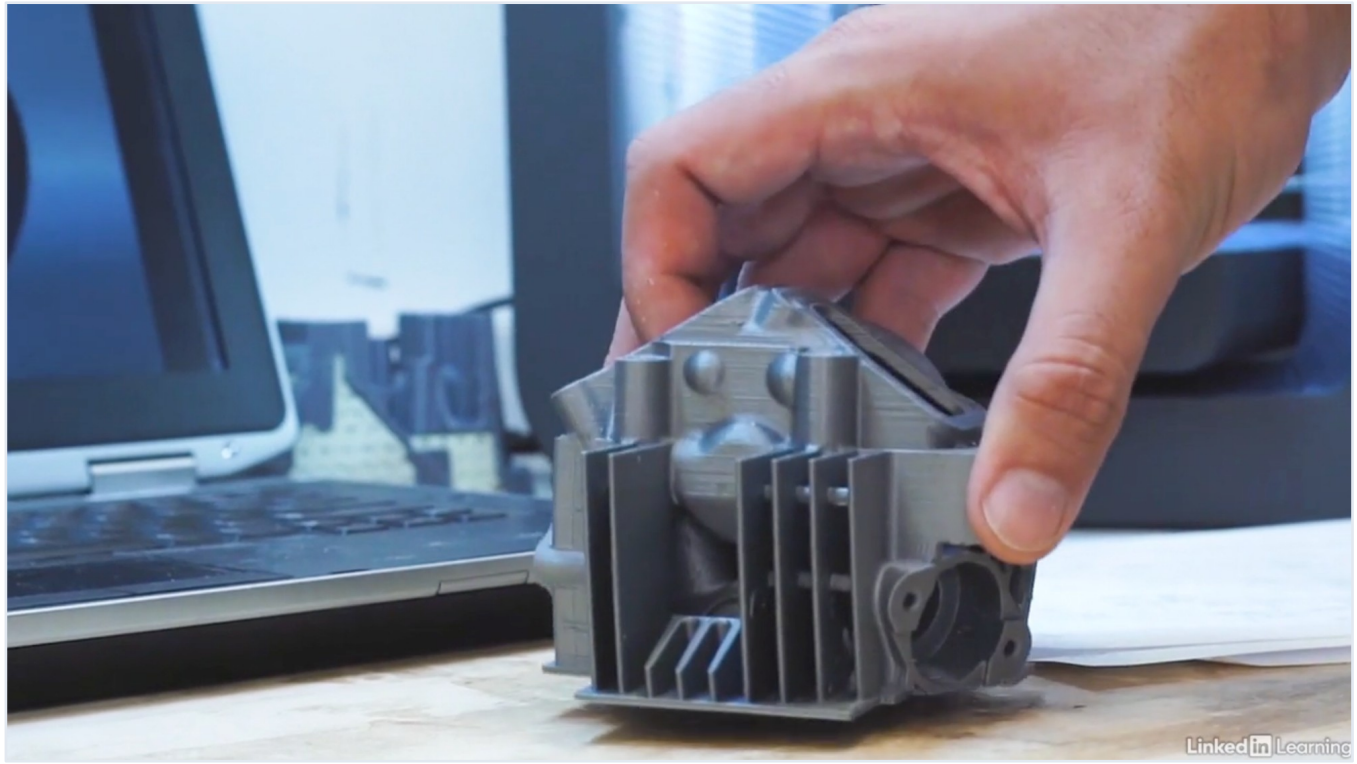


A dog fitted with 3D-printed prosthetic legs

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CASE STUDY · K4

Case Study — Distributed Manufacturing



A replacement part printed where it is needed

Ship files, not parts

Physical goods distributed as downloadable files — like music and movies.

Print at point of need

Spare parts printed on-site cut inventory, logistics and lead time.

Real constraints

Quality and materials remain limited; printers need calibration and maintenance.

People still matter

Staff must be trained, filament supplied, tech support provided.

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K4 · WHO USES IT

AM Across Industries

1

Aerospace

Lightweight blisks, brackets and fuel nozzles — mass down, lead time down.

2

Automotive

Rapid prototypes, custom jigs & fixtures, low-volume spares.

3

Medical & dental

Patient-specific implants, surgical guides, aligners, hearing aids.

4

Rail & heavy industry

Bespoke obsolete spares and worn-part repair by DED.

5

Construction

Extruded-concrete walls and printed architectural elements.

6

Consumer & education

Desktop prototyping, maker culture, STEM classrooms.

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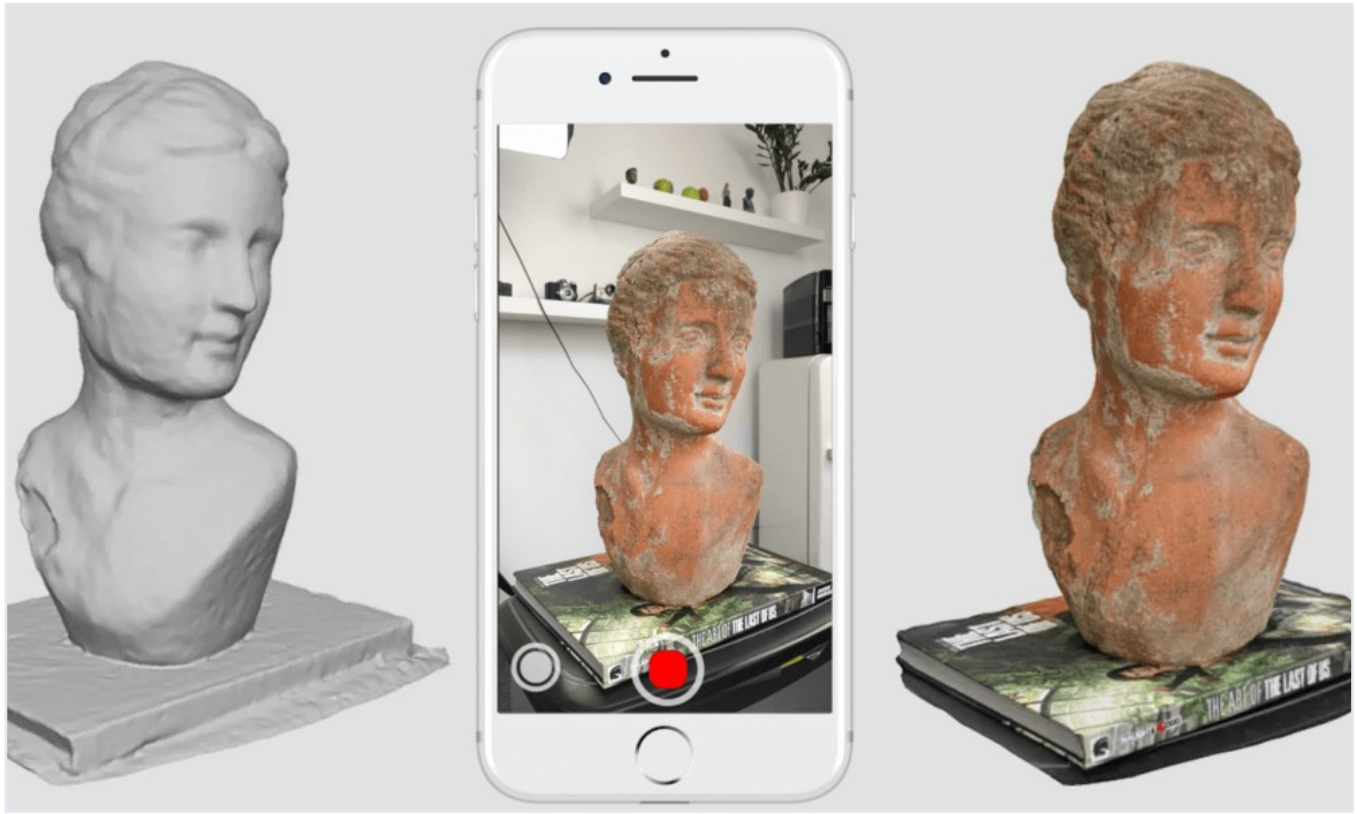
THE DELOITTE LENS · K4/K6

Complexity is free. Customisation is free. Volume is not.

AM rewrites the economics of scope and scale — prototype and personalise with AM; mass-produce with molds.

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3D Scanning — Capture Reality



Photogrammetry: phone capture → reconstructed bust

Photogrammetry
Dozens of photos from all angles reconstructed into a textured 3D mesh.

LiDAR / structured light
Depth sensors on phones and dedicated scanners capture geometry directly.

Mobile apps
Polycam, Kiri Engine, Scandy Pro, Heges, ItSeez3D, EM3D, Trnio, Capture 3D.

Output
A mesh (STL/OBJ) ready for repair, remodelling and re-printing.

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TOOLBOX

3D Model Software Landscape

1. **TinkerCAD**
   Free browser CAD — drag-and-drop solids, perfect first step.

2. **Fusion 360**
   This course's tool — parametric sketch-driven CAD/CAM with free personal licence.

3. **Inventor / SolidWorks**
   Professional parametric MCAD for production engineering.

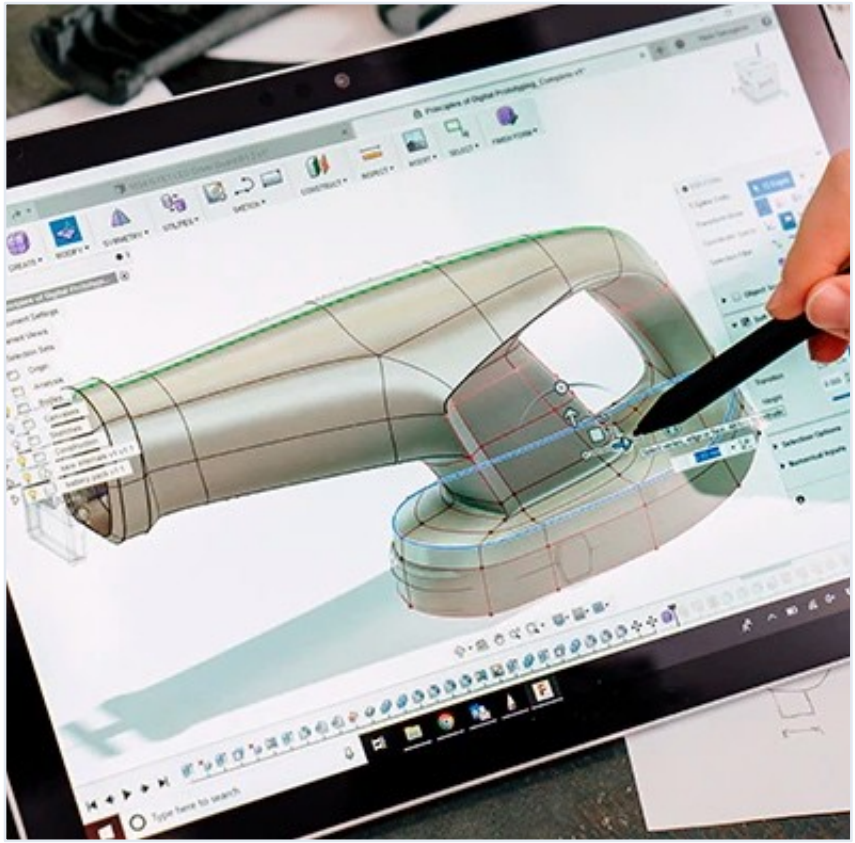
4. **Onshape**
   Full cloud CAD — CAD in a browser tab, built-in PDM.

5. **SketchUp**
   Fast push-pull architectural modelling.

6. **GrabCAD / Thingiverse / YouMagine**
   Model repositories — millions of ready-to-print designs.

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Fusion 360 — Sketch-Driven CAD



Fusion 360 modelling a consumer product enclosure

Sketch first

2D shapes drawn on sketch planes become 3D parts through modelling operations.

Dimension-accurate

Parametric dimensions drive printable, engineering-accurate geometry.

Four core techniques

Extrude, Revolve, Sweep and Loft cover most printable parts — Labs 1–4.

Beyond solids

Freeform (T-spline) and surface modelling handle organic, complex curvature.

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Fusion 360 — The Four Core Techniques

1

Extrude — Lab 1

Draw a 2D profile, push it along the third axis into a solid — the bread-and-butter operation.

2

Sweep — Lab 2

Move a profile along a 2D path on a perpendicular plane — tubes, clips, wires.

3

Revolve — Lab 4

Spin a half cross-section around an axis — funnels, vases, pulleys, bottles.

4

Loft — Lab 3

Blend between profiles on different planes, guided by rails — freeform vessels and transitions.

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Freeform and Surface Modelling

- Freeform (T-spline) modelling shapes organic forms by moving, rotating and scaling the points, edges and faces of primitive bodies.
- Surface modelling constructs complex curvature patch by patch, then thickens or stitches into printable solids.
- Use them when sketch-based solids cannot reach the shape — ergonomic grips, sculpted housings, scanned-form matching.
- Lab 6's scanned meshes and the torso case study are edited with exactly these tools.

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DESIGN QUALITY · A1

Designs with Stress Concentration

1

What it is

A point where local stress runs far higher than the surrounding material.

2

Where it hides

Sharp corners, holes, notches and grooves — any sudden change in geometry.

3

Why AM cares

Layer boundaries + stress risers = the crack starts exactly there.

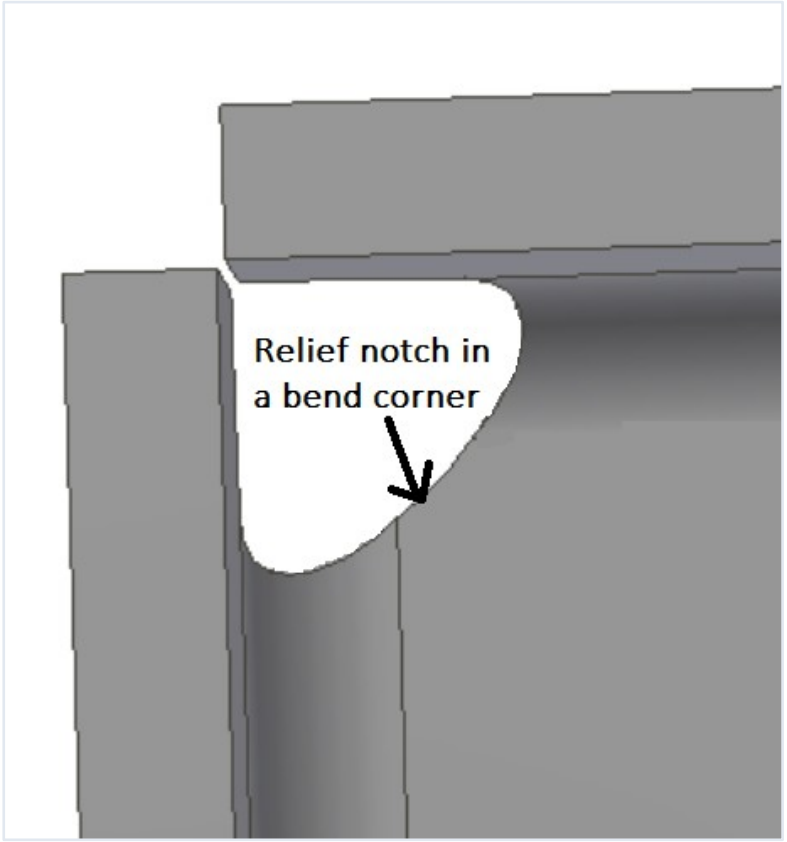
4

The fix

Fillet radii, tapers and relief notches ease the transition and spread the load.

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Design Tips to Minimise Stress Concentration



A relief notch in a bend corner de-concentrates stress

Fillet every corner

Use fillet radii or tapers to ease transitions between shapes.

Relief notches

A round relief notch stops a crack from starting in a sharp bend.

Avoid sharp corners

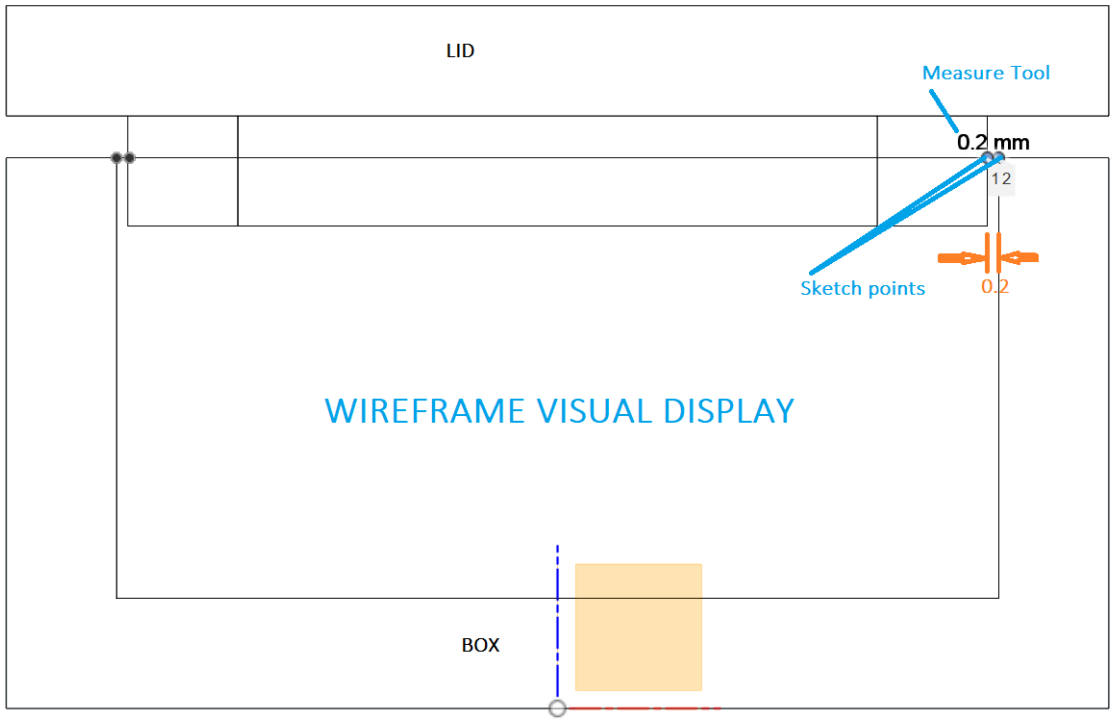
Especially at load-bearing intersections and layer boundaries.

Orient for strength

Put tensile loads along the layers, not across them.

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Digital and Physical Tolerance



Wireframe visual display of a lid-to-box clearance measurement

Design the gap

Assembling parts need designed clearance — 0.2 mm is the FDM starting point.

Measure digitally

Fusion 360's Measure, Section Analysis and sketch tools verify the designed gap.

Verify physically

Digital calipers confirm the printed part reproduced the digital tolerance.

Lab 5

You'll design, print and measure a 0.2 mm container-and-lid fit.

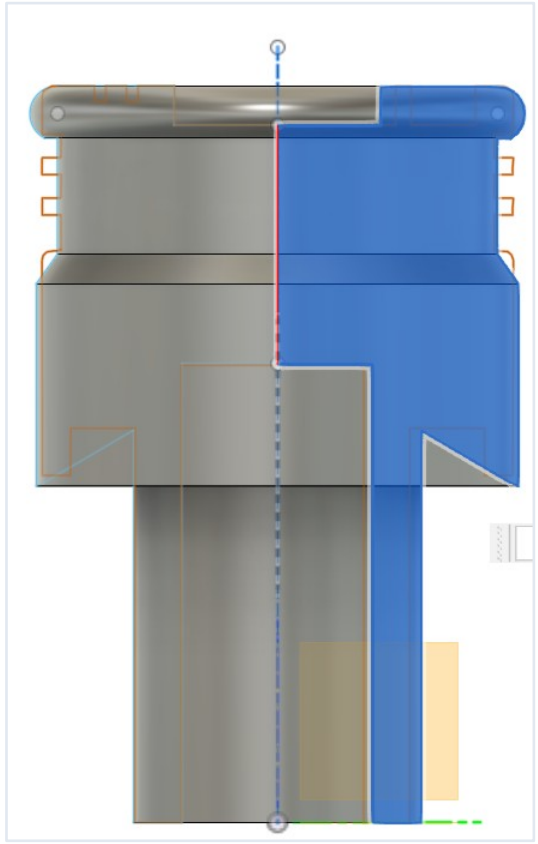
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Reverse Engineering with 3D Scanning

1. IDENTIFY — inspect and repair damage on the original part
2. SCAN — capture it with a high-resolution 3D scanner
3. REFINE — clean the mesh into an accurate CAD model
4. SELECT — pick the AM process for the material & design
5. MANUFACTURE — print and post-process the component

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Case Study — Reverse Engineering a Mesh



Sectioning a scanned piston mesh to recover its true profile

Import the scan

A ready-made mesh/scan imports straight into Fusion 360.

Section the mesh

The mesh section tool produces a clean cross-section curve.

Trace & rebuild

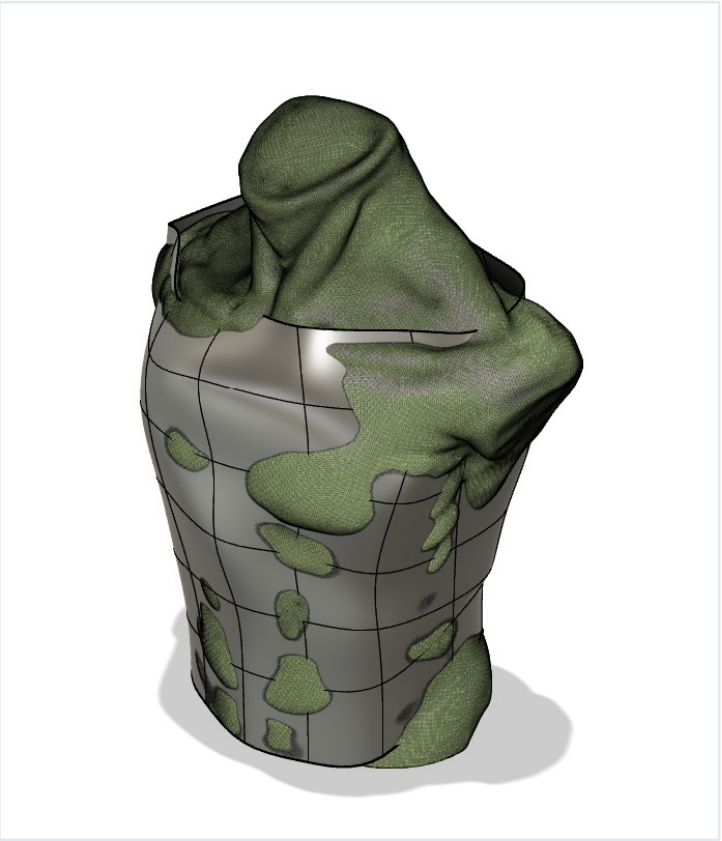
Trace the section with a corrected sketch — fix wear and damage digitally.

Mirror & revolve

Rebuild the part as clean parametric CAD derived from the scan.

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Case Study — Reverse Engineering an Organic Form



T-spline body matched to a torso scan, ready to thicken

Freeform cylinder
Start a freeform cylinder over the torso scan and match its contours.

Pull to the scan
The Pull tool snaps the cylinder's vertices onto the scanned surface.

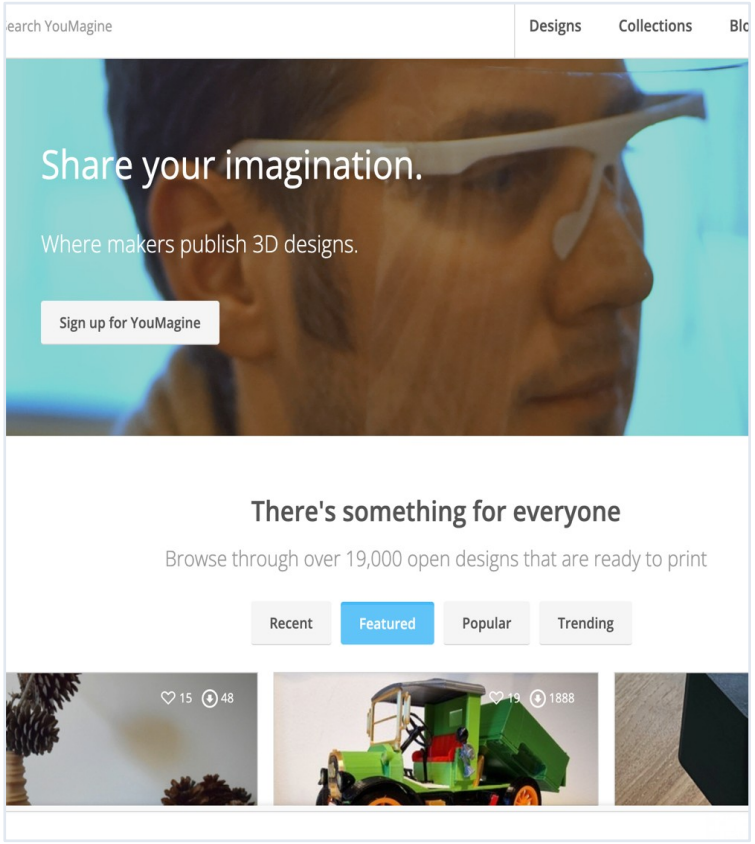
Thicken
Thicken the matched surface into a printable, wearable shell.

Result
Custom-fit body armour conformal to the scanned torso.

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READY-MADE MODELS

3D Printing Repositories



Browse, download, slice — millions of proven printable designs

Thingiverse
thingiverse.com — the largest free STL library.

YouMagine
youmagine.com — Ultimaker's open design community.

Printables / GrabCAD
Curated models and engineering-grade CAD downloads.

Check the licence
Respect Creative Commons terms before commercial use.

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WHAT YOU'LL DO · LABS 1–6

Hands-On Labs — Topic 1

Design for AM

- Lab 1: Extrude — keychain token
- Lab 2: Sweep — paper clip
- Lab 3: Loft — plant pot

Design precision

- Lab 4: Revolve — funnel
- Lab 5: Tolerance — container & lid

Capture reality

- Lab 6: 3D scan a real object
- Export, repair and print the scan

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TOPIC 01 · HANDS-ON

Lab 1 — Extrude — Personalised Keychain Token

LAB 1

A1 — determine material and design considerations while modelling a printable part with the extrude technique

Model an elliptical keychain token in Fusion 360, personalise it with the first letter of your name, then slice it in Cura and 3D print it.

You'll build

A personalised elliptical keychain token, sliced and 3D printed in PLA.

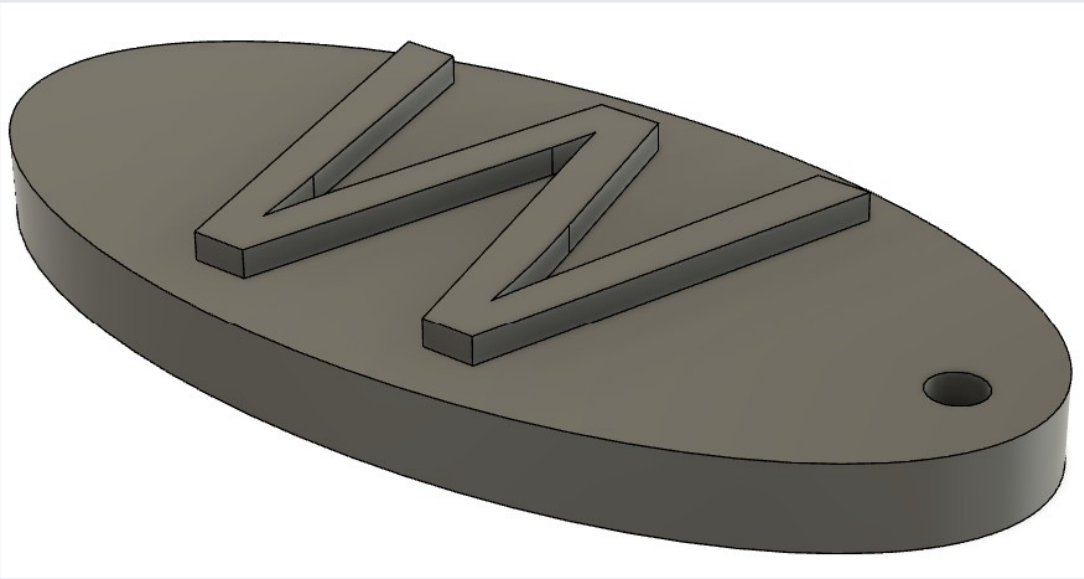
Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

Step-by-step guide: Learner Guide, Lab **1** · labs/lab-01 folder

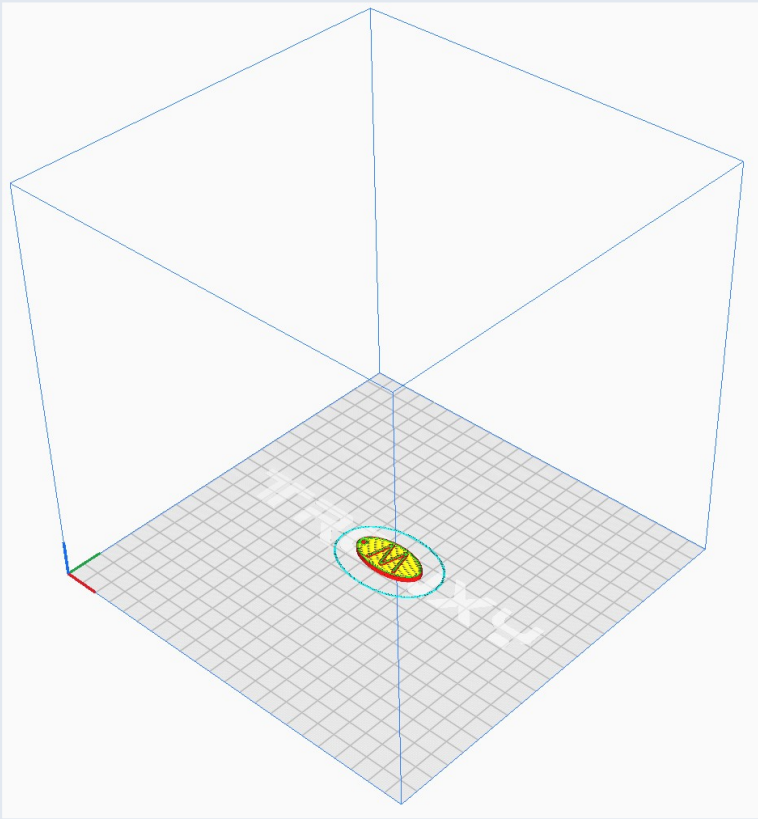
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TOPIC 01 · HANDS-ON

Lab 1 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 1 · VERIFY

Lab 1 — Extrude — Personalised Keychain Token

✅ Test it

The sliced preview shows the token printing flat with no supports; the printed token carries your raised letter and the keyring hole is clear.

Full steps and troubleshooting: Learner Guide · Lab **1**

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TOPIC 01 · HANDS-ON

Lab 2 — Sweep — Giant Paper Clip Bookmark

LAB 2

A1 — apply the sweep technique to produce a printable path-driven part

Model a giant paper clip bookmark in Fusion 360 by sweeping a circular profile along a paper-clip path, then slice and 3D print it.

You'll build

A giant paper clip bookmark modelled with the sweep technique and 3D printed flat.

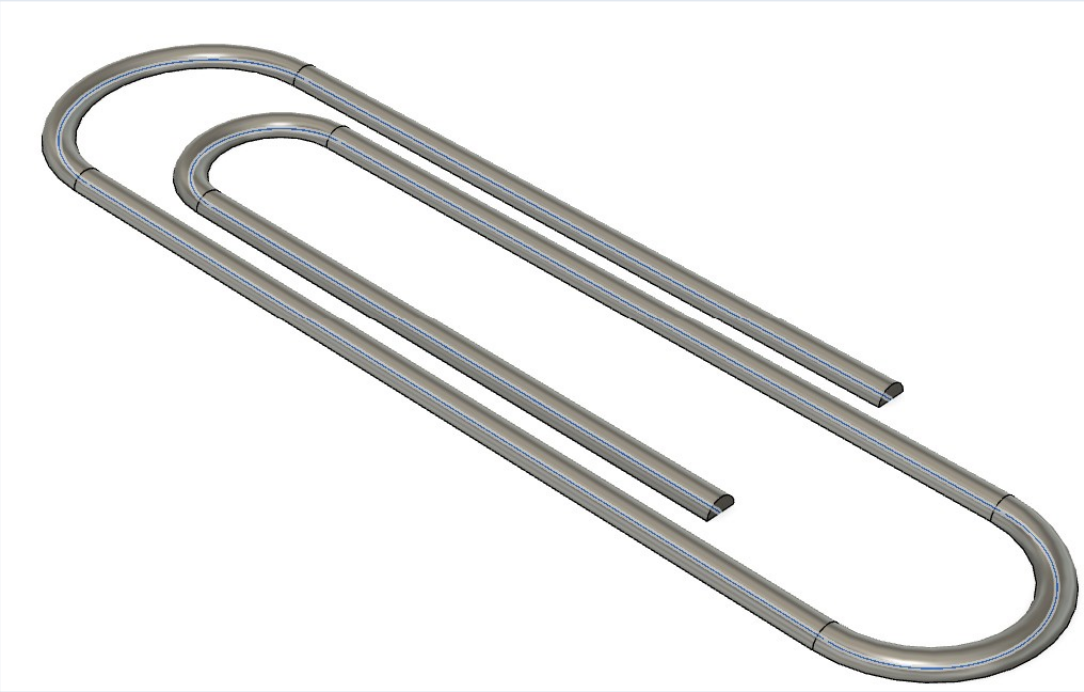
Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

Step-by-step guide: Learner Guide, Lab **2** · labs/lab-02 folder

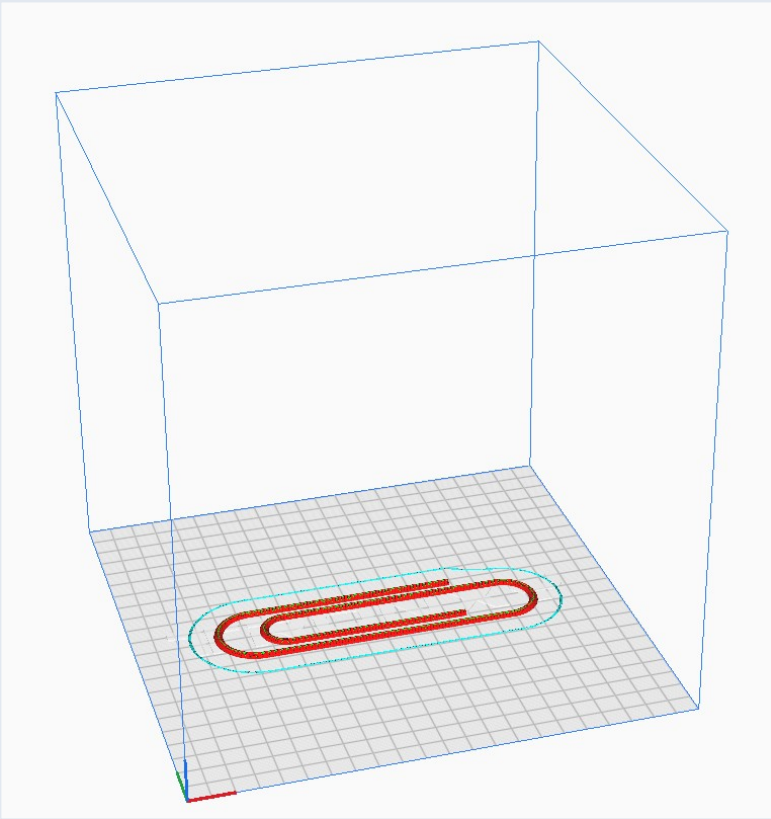
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TOPIC 01 · HANDS-ON

Lab 2 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 2 · VERIFY

Lab 2 — Sweep — Giant Paper Clip Bookmark

✅ Test it

The preview shows the whole clip printing in contact with the bed; the printed clip flexes slightly and grips a few pages like a real paper clip.

Full steps and troubleshooting: Learner Guide · Lab **2**

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TOPIC 01 · HANDS-ON

Lab 3 — Loft — Mini Plant Pot with Custom Rails

LAB 3

A1 — apply the loft technique with guide rails to create a freeform printable vessel

Model a mini plant pot in Fusion 360 by lofting between a square top and a circular base with custom guide rails, then slice and 3D print it.

You'll build

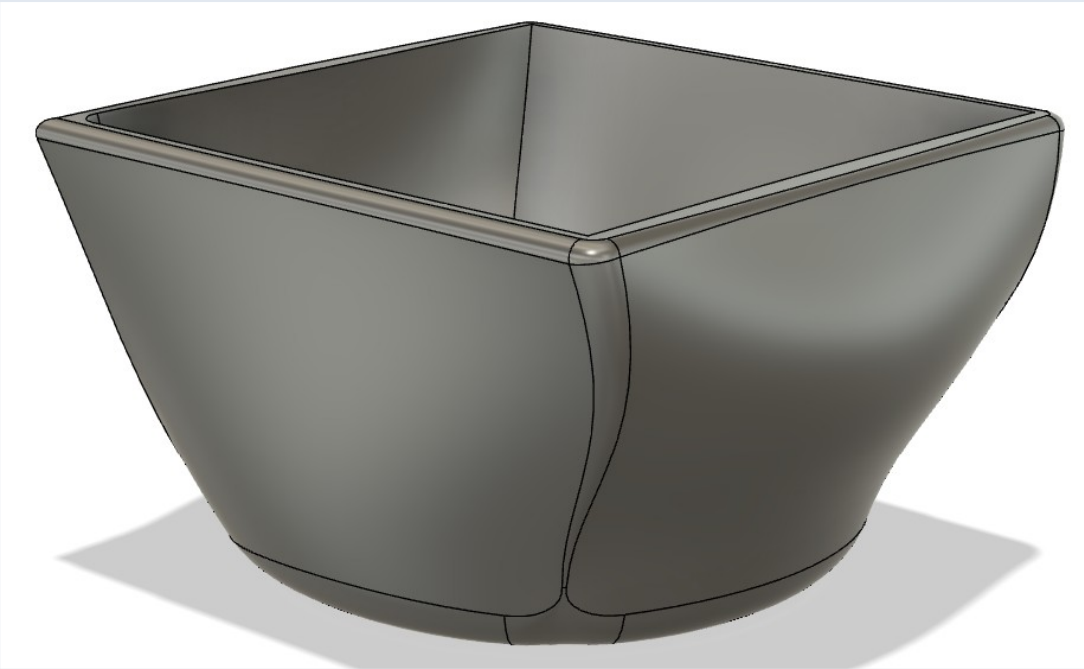
A mini plant pot lofted from circle to square, 3D printed without supports.

Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

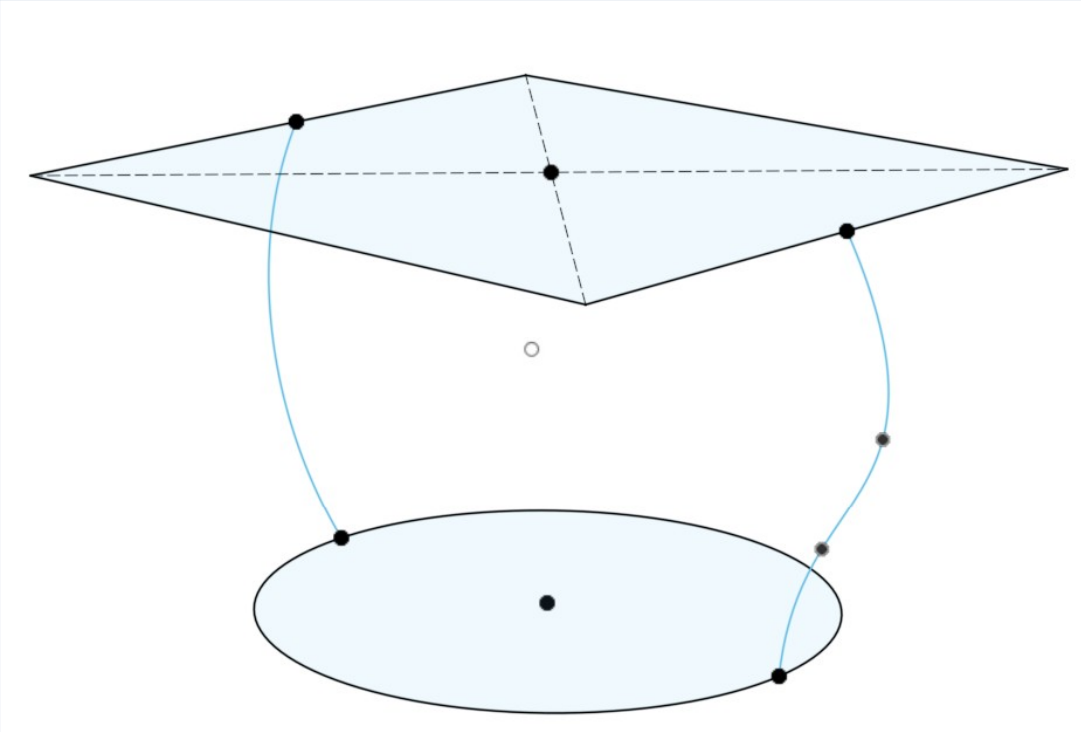
Step-by-step guide: Learner Guide, Lab **3** · labs/lab-03 folder

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Lab 3 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 3 · VERIFY

Lab 3 — Loft — Mini Plant Pot with Custom Rails

✅ Test it

The pot prints support-free with an even 2.5 mm wall; the loft transition from circle to square is smooth with no layer separation.

Full steps and troubleshooting: Learner Guide · Lab **3**

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TOPIC 01 · HANDS-ON

Lab 4 — Revolve — Funnel from a 2D Profile

LAB 4

A1 — apply the revolve technique and choose a print orientation that avoids supports

Model a funnel in Fusion 360 by sketching its half cross-section and revolving it about the vertical axis, then invert it in Cura and 3D print it.

You'll build

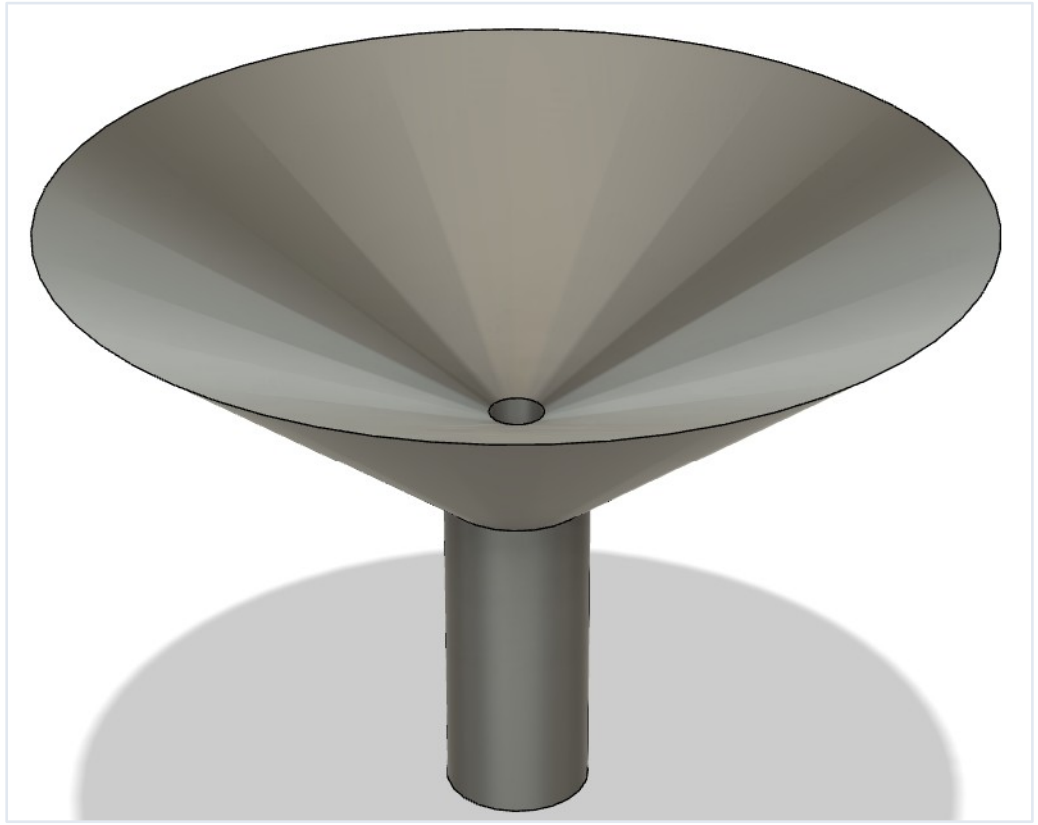
A working funnel revolved from a single 2D profile, printed wide-end down.

Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

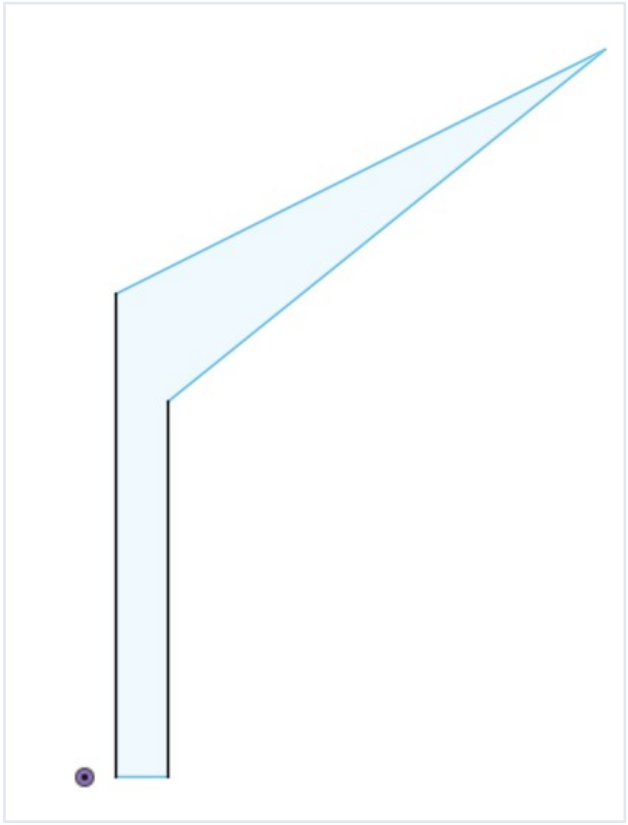
Step-by-step guide: Learner Guide, Lab **4** · labs/lab-04 folder

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Lab 4 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 4 · VERIFY

Lab 4 — Revolve — Funnel from a 2D Profile

✅ Test it

Sliced upside-down the funnel needs no supports; water poured into the printed funnel flows cleanly through the spout.

Full steps and troubleshooting: Learner Guide · Lab **4**

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TOPIC 01 · HANDS-ON

Lab 5 — Tolerance — Container and Lid (0.2 mm Fit)

LAB 5

A1 — determine digital and physical tolerances between assembling parts

Model a container and lid as two separate bodies in Fusion 360 with a 0.2 mm clearance, verify the digital tolerance, then 3D print and test the physical fit.

You'll build

A two-part container whose lid slides on with a designed 0.2 mm clearance.

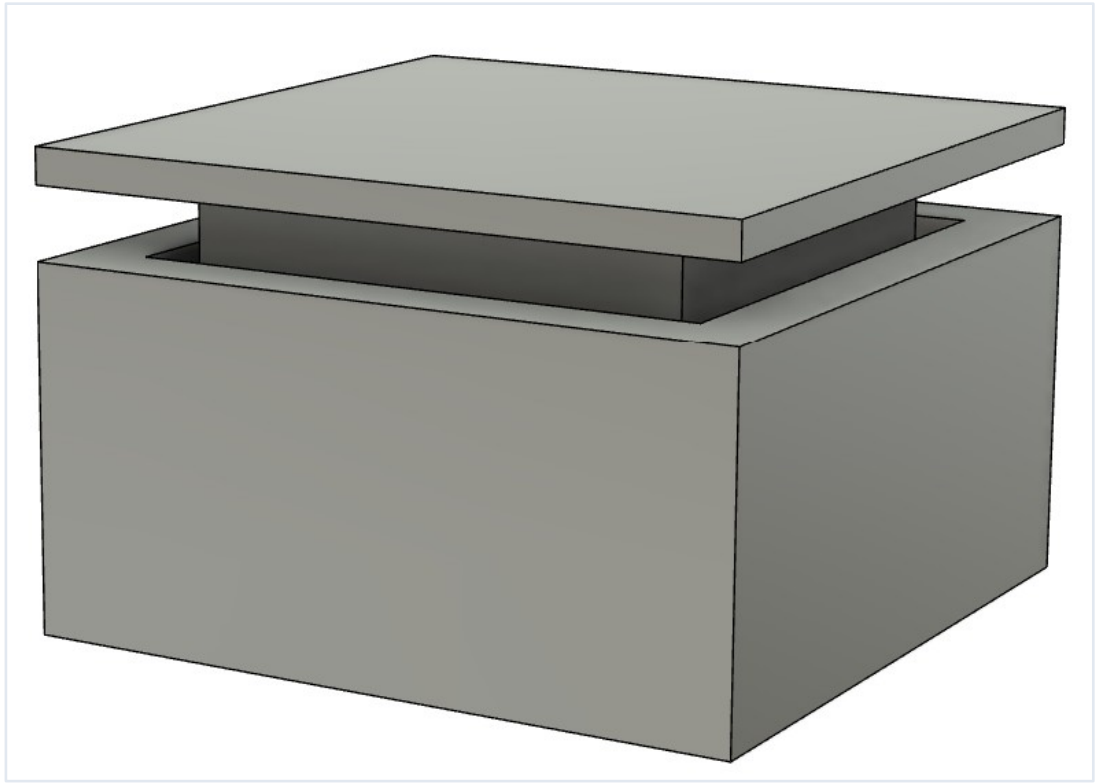
Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA), digital calipers, supplied FDM Box reference files

Step-by-step guide: Learner Guide, Lab **5** · labs/lab-05 folder

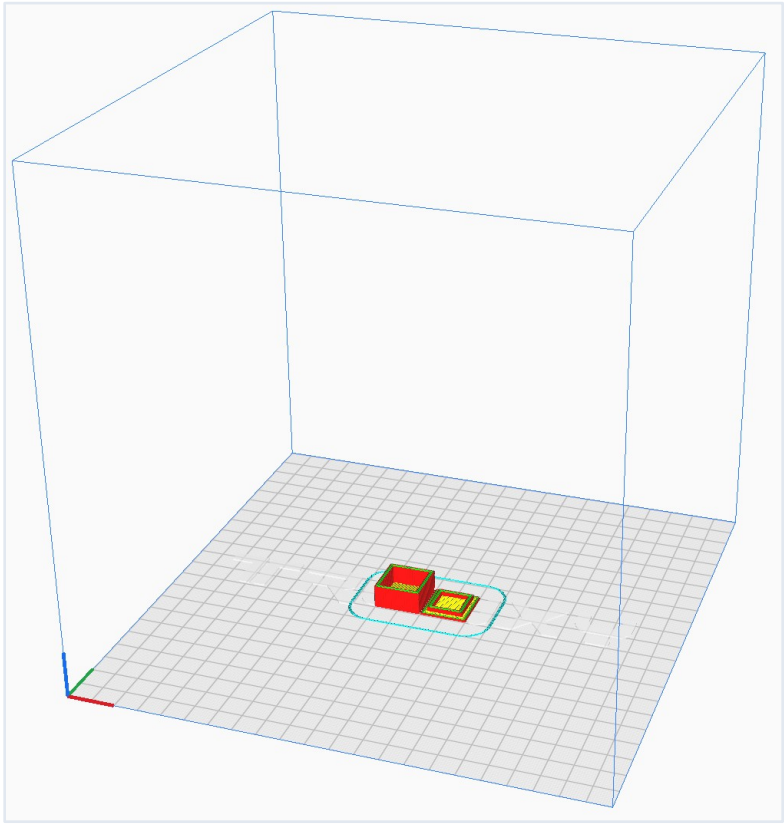
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TOPIC 01 · HANDS-ON

Lab 5 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 5 · VERIFY

Lab 5 — Tolerance — Container and Lid (0.2 mm Fit)

✅ Test it

The lid slides onto the container with light friction — neither loose nor forced — and your evidence distinguishes the designed 0.2 mm clearance, the measured printed result and the supplied FDM Box reference setup.

Full steps and troubleshooting: Learner Guide · Lab **5**

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TOPIC 01 · HANDS-ON

Lab 6 — 3D Scan and Print a Real Object

LAB 6

A4 — evaluate reverse engineering of a component through 3D scanning and subsequent AM

Scan a real object with a mobile 3D-scanning app, export the mesh as an STL file, repair it if needed, then slice and 3D print the reproduction.

You'll build

A 3D-printed reproduction of a real object captured by photogrammetry on your phone.

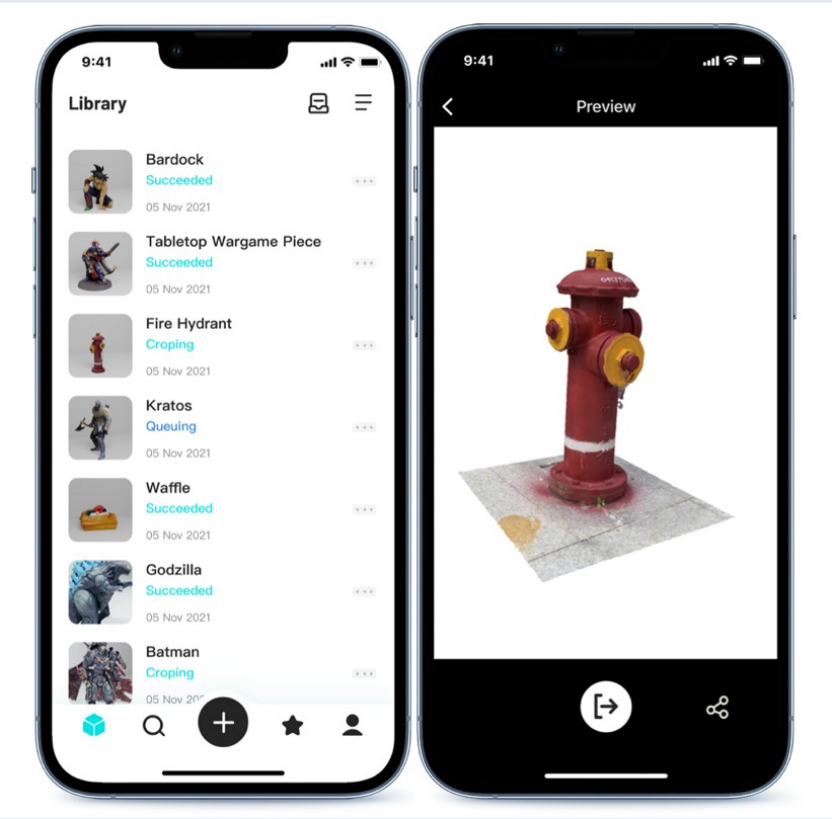
Tools: Polycam / Kiri Engine / Scandy Pro (mobile), Fusion 360, Ultimaker Cura

Step-by-step guide: Learner Guide, Lab **6** · labs/lab-06 folder

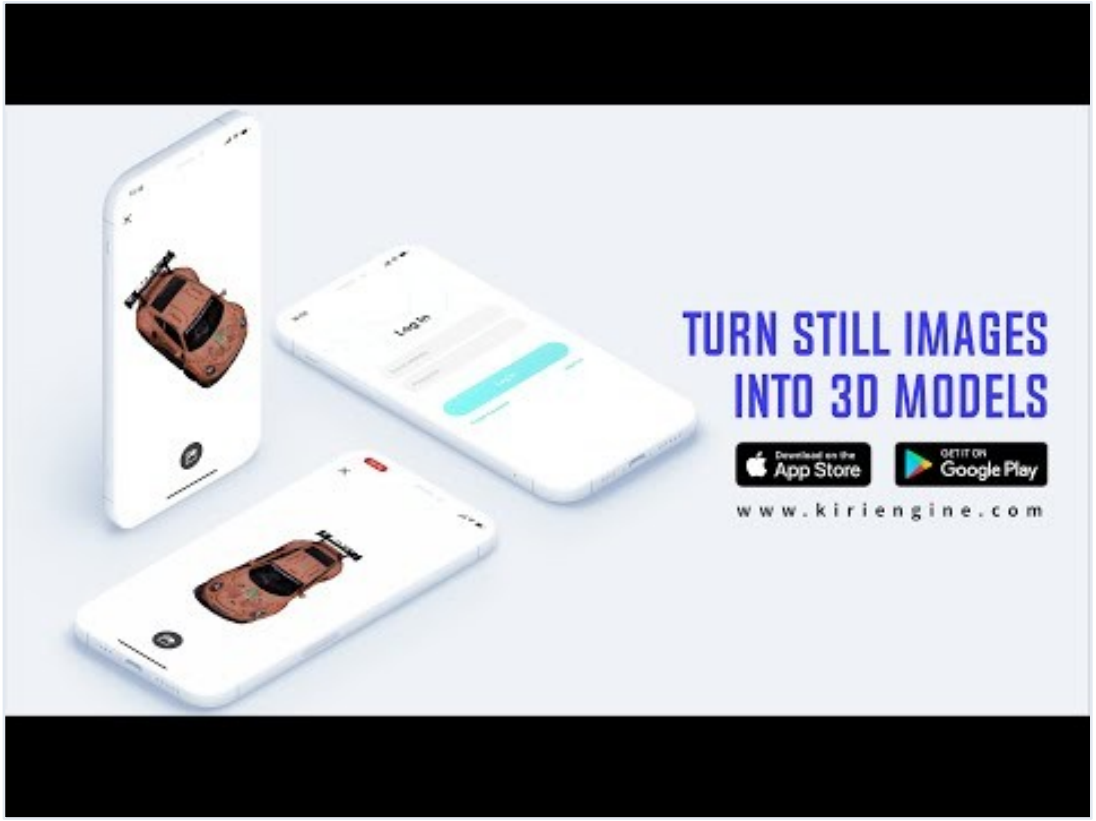
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TOPIC 01 · HANDS-ON

Lab 6 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 6 · VERIFY

Lab 6 — 3D Scan and Print a Real Object

✅ Test it

The exported STL loads into Cura as a watertight mesh, and the printed part is a recognisable scaled reproduction of the scanned object.

Full steps and troubleshooting: Learner Guide · Lab **6**

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TOPIC 01 · RECAP

Recap — Topic 1

- AM builds parts layer by layer from CAD with no part-specific tooling — seven process families cover polymers to metals.
- Material, complexity, volume, post-processing and cost drive the selection of an AM solution.
- AM's benefits (freedom, speed, customisation, low waste) trade against throughput, anisotropy and finishing.
- 3D scanning + Fusion 360 turn real objects into editable CAD for re-manufacture — reverse engineering.
- You modelled with extrude, sweep, loft and revolve, engineered a 0.2 mm fit, and printed a scanned object.

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02

TOPIC 02

Process Related Issues for AM

Printer hardware · slicing · adhesion · extrusion · temperature · materials (K2, K4 · A2, A3)

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FIVE SOURCES · K2

Where AM Issues Come From

1

Printer hardware

Frame, motion system, extruder, hotend, bed — wear and misalignment show up in every print.

2

Materials

Diameter variation, moisture, contamination and wrong temperature windows.

3

Printing process conditions

Speed, temperature, retraction, cooling and adhesion settings.

4

Environment conditions

Draughts, ambient temperature, humidity and vibration.

5

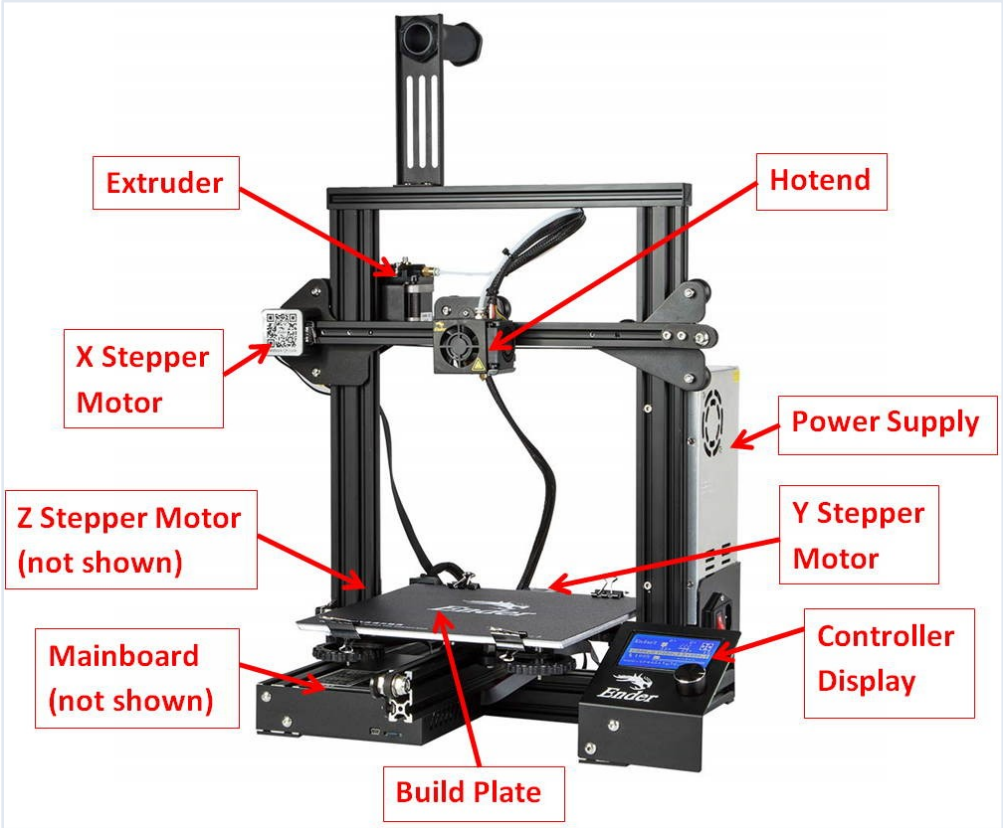
Design-caused

Overhangs, thin walls, sharp corners and tolerance choices that fight the process.

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HARDWARE · K2

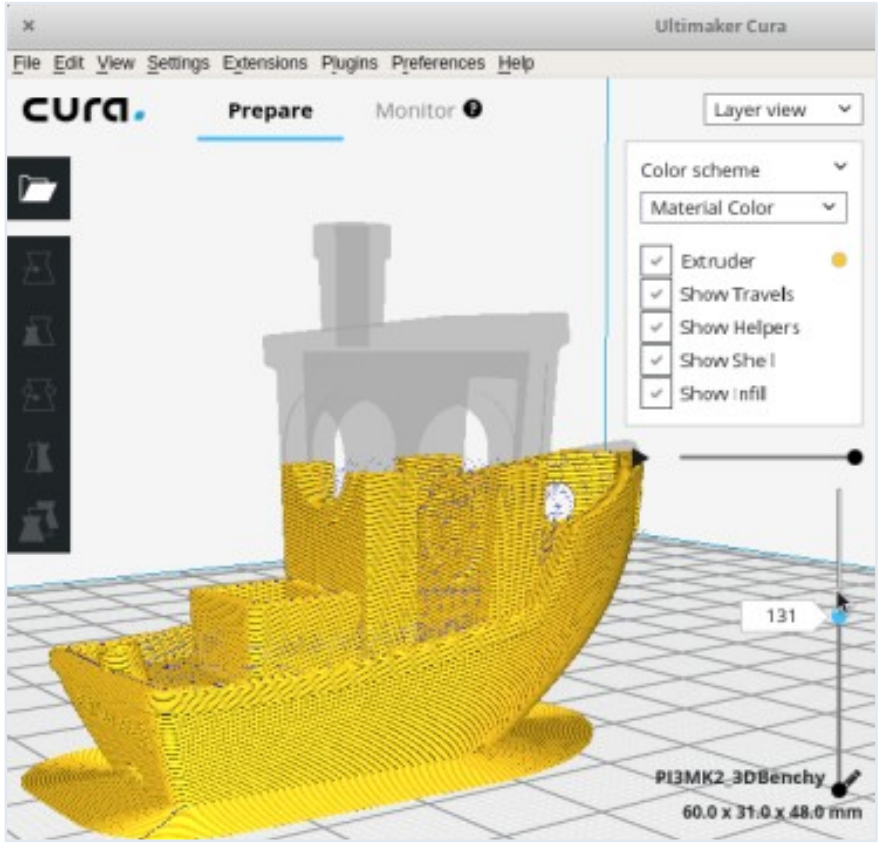
3D Printer Anatomy



Extruder · hotend · X/Y/Z steppers · build plate · power supply · mainboard · controller — know the part to diagnose the fault

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The Slicer — From Mesh to G-code



Cura preparing a hull model — layer preview of the sliced toolpath. Alternatives: PrusaSlicer, SelfCAD, OctoPrint, IceSL, KISSlicer

What it does

Converts the 3D model (STL) into specific layer-by-layer instructions for the printer.

G-code

The output is G-code — movement, temperature and extrusion commands the printer executes.

Settings = quality

Every quality, speed and strength decision is made here, before the printer moves.

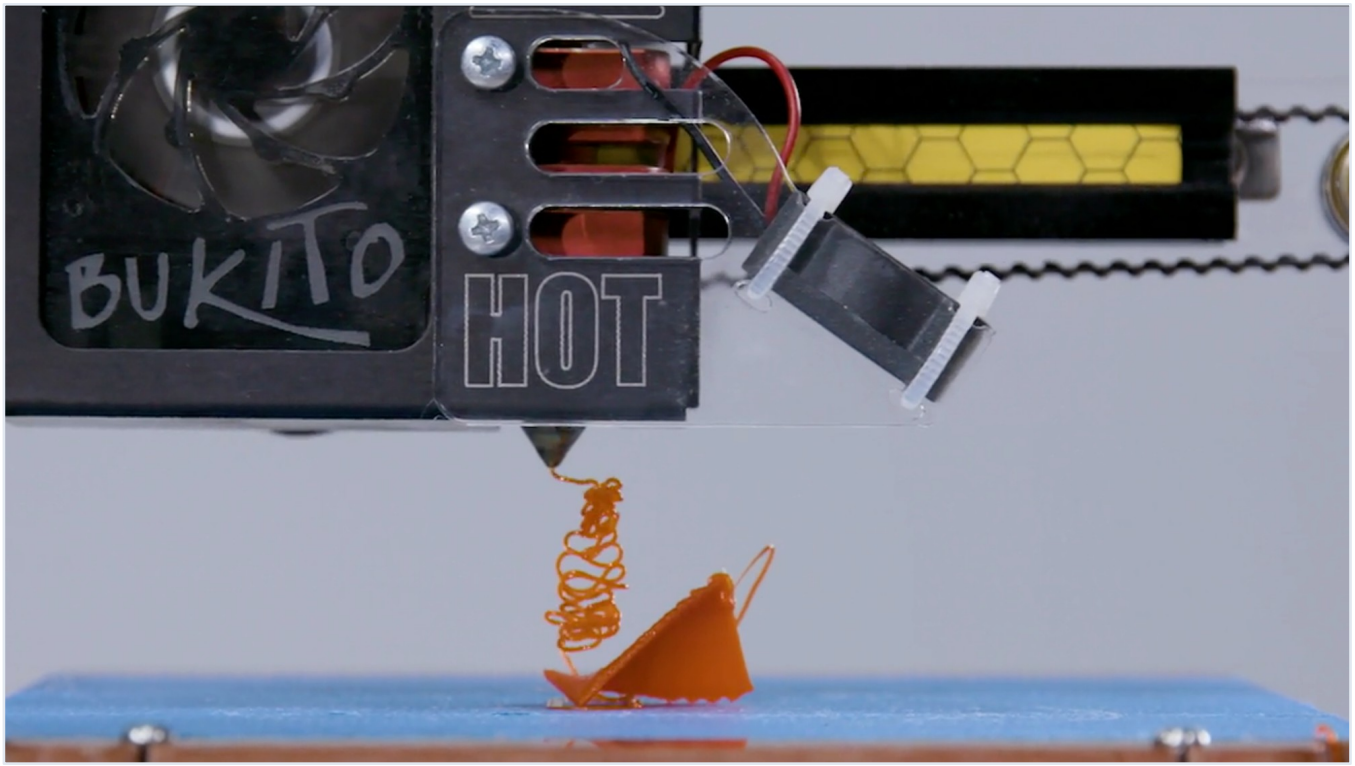
Cura

This course uses Ultimaker Cura — free, powerful, industry-standard for FDM.

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FIRST LAYER · K2/A2

Getting the Print to Stick



An adhesion failure mid-print — the part broke loose and dragged

Print geometry

Small or complex bases have little contact area to grip the bed.

Nozzle distance

First-layer gap must be right — too far and nothing sticks, too close and it drags.

Platform surface

Bed material and films/coatings must be compatible with your filament.

Platform temperature

A heated bed keeps the first layers relaxed and attached.

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Raft, Brim and Skirt

Raft

- A printed grid platform the model sits on
- Rescues warping-prone and top-heavy parts
- Harder to remove; roughens the bottom face

Brim

- A flat border fused to the part's first-layer edge
- Boosts adhesion and fights warping
- Peels away; light cleanup at the edge

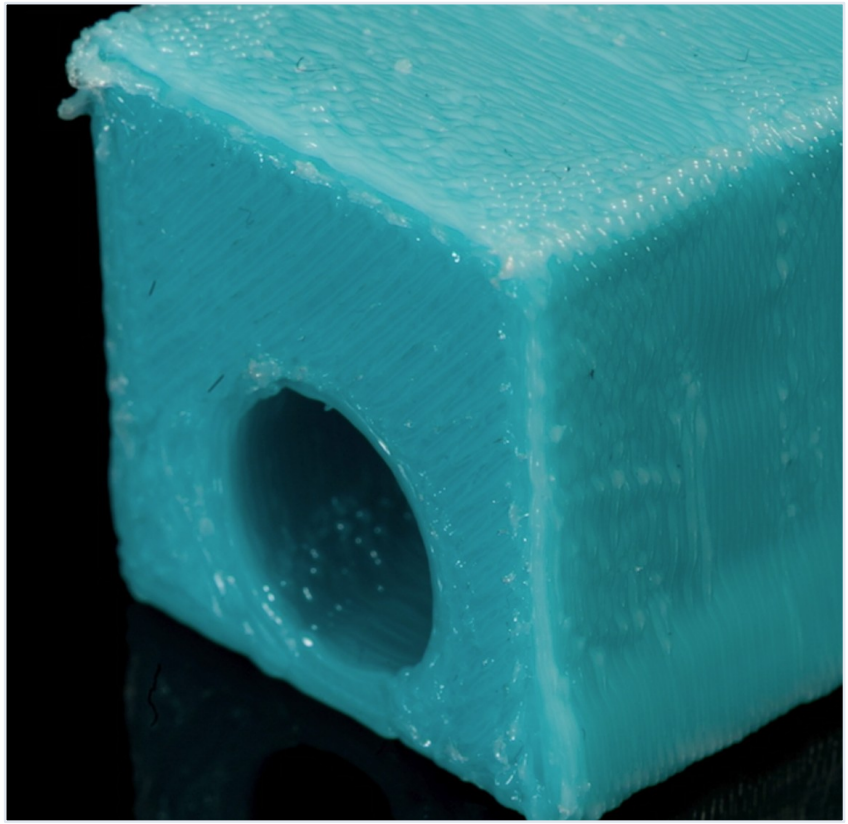
Skirt

- Loops around the part, never touching it
- Primes the nozzle, evens the flow
- Doubles as a bed-levelling check

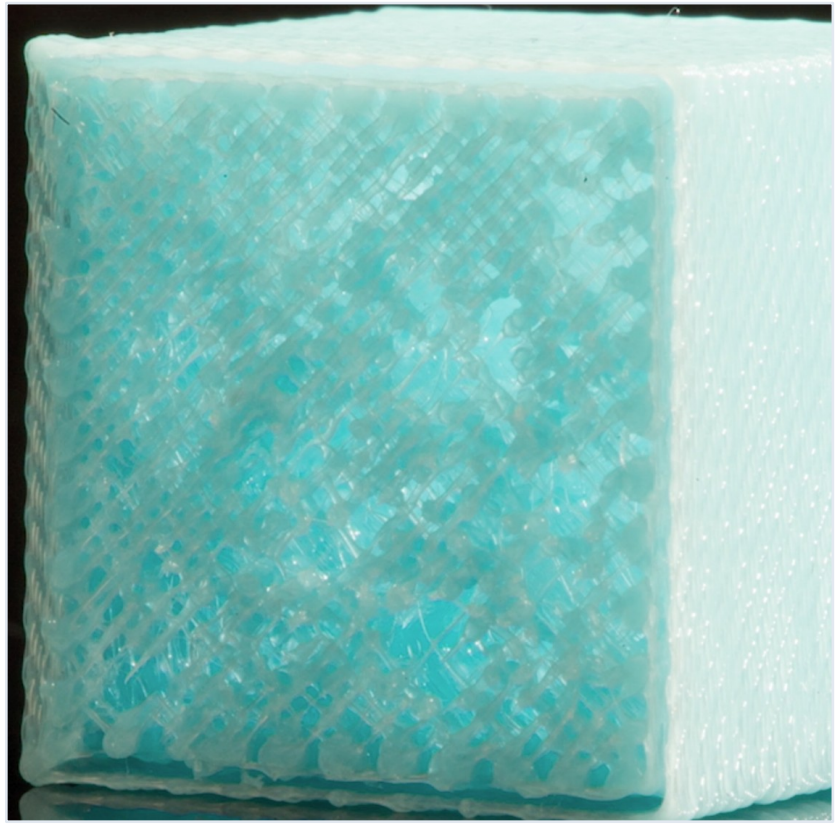
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DIAGNOSIS · A2

Extrusion Issues — Over vs Under



Over-extrusion: blobby surfaces, oversized walls, ridged tops



Under-extrusion: gaps, thin walls, missing layers

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A2/A3 · CAUSE → FIX

Diagnosing Extrusion Faults

Too much

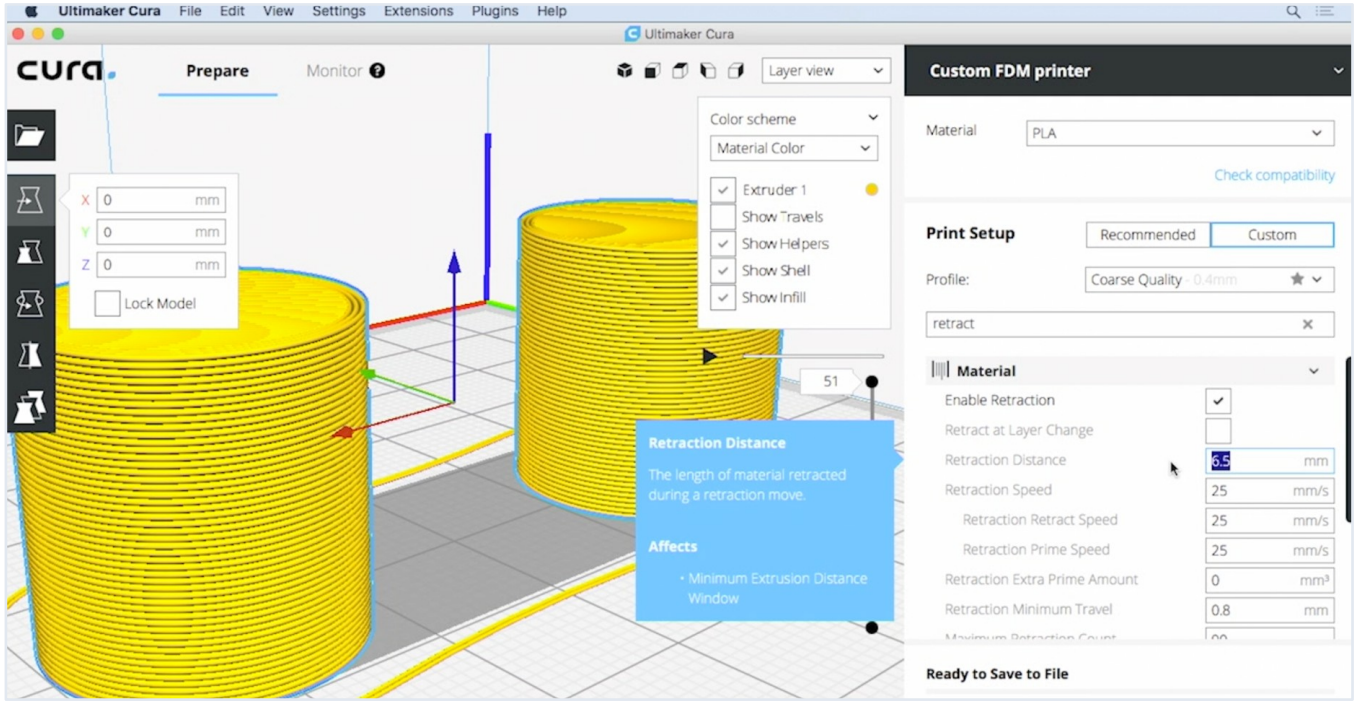
- Over-extrusion — too much plastic
  - Blobby prints; holes come out undersized
  - Ridges stick up from top surfaces
  - Fix: calibrate flow %, check filament diameter
  - Fix: lower temperature a step

Too little

- Under-extrusion — too little plastic
  - Gaps between lines; weak, brittle layers
  - Extruder clicking or grinding the filament
  - Fix: check for partial clogs and jam points
  - Fix: raise temperature / slow down / dry filament

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Retraction — Stopping the Flow



Cura retraction settings — distance and speed

Why retract

Plastic in the nozzle is under pressure; jumps across open space would leak.

The move

The extruder pulls filament back a few millimetres to equalise pressure.

Distance & speed

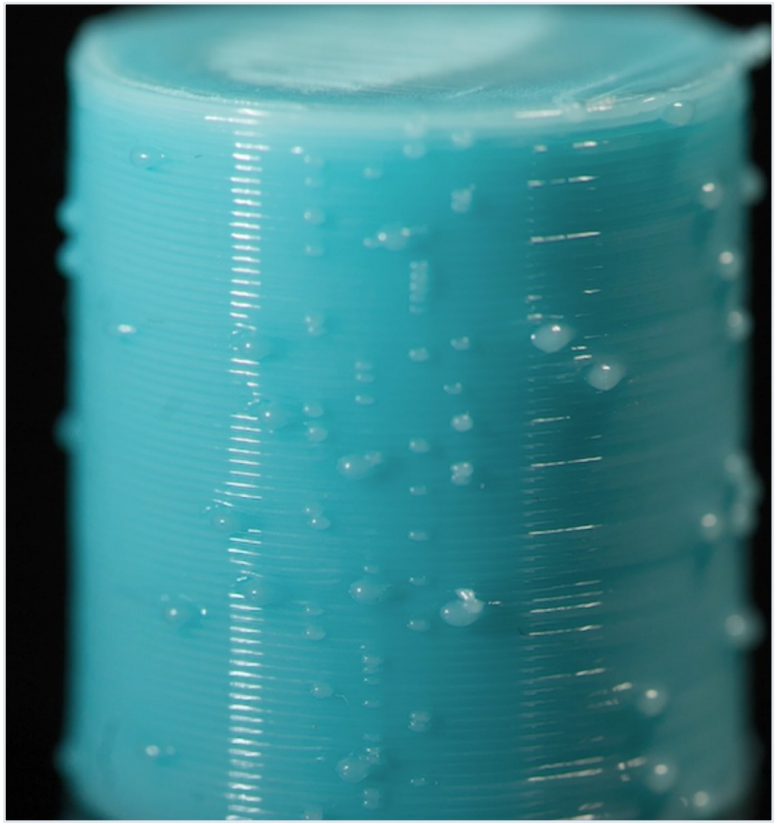
Tuned per printer/material — Bowden setups need more than direct drive.

Get it wrong

Too little → oozing and strings; too much → dimples and jams.

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Oozing and Stringing



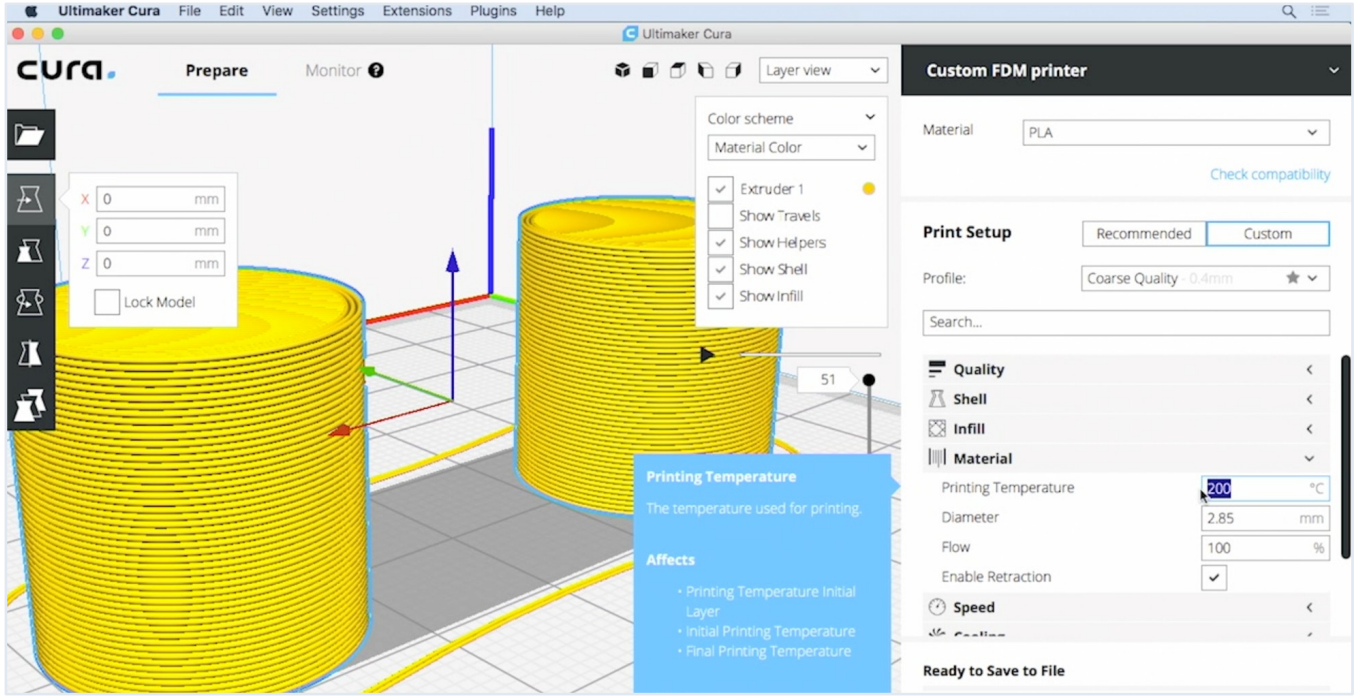
Blobs and dimples where pressure equalised badly



Strings bridge the travel moves between towers

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Temperature — The Master Variable



Setting printing temperature in Cura's material section

Too low — way
Extrusion fails: clicking extruder, ground filament, nothing flows.

Too low — slightly
Layers don't bond; the part delaminates along layer lines.

Too high — slightly
Bridges and overhangs sag; stringing hard to control.

Too high — way
Smoking, discolouration, dripping liquid instead of controlled extrusion.

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A2 · EVIDENCE

Reading Temperature Faults



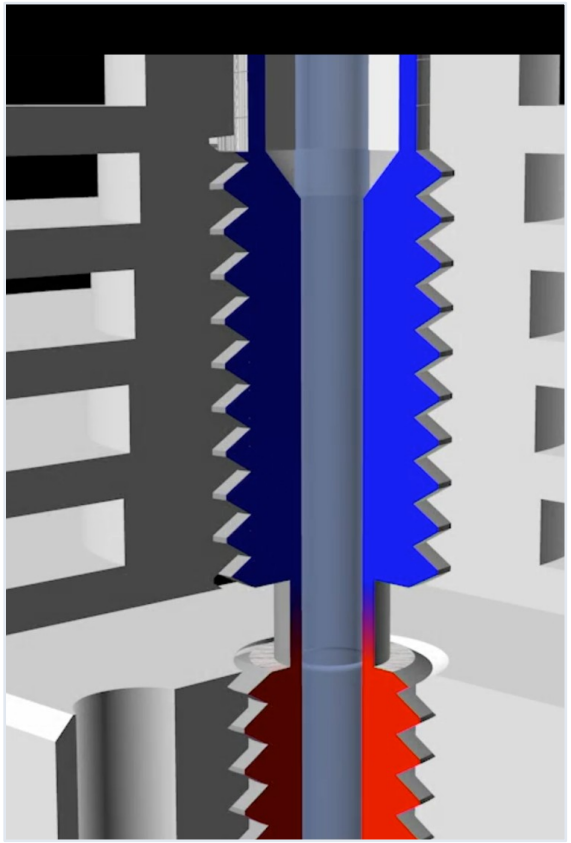
Bonding too cold — layers separate under light force



Far too hot — sagging, melted detail, discolouration

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Filament Jams and Heat Creep



Heat creep: the melt zone (red) spreading up the cold side

The jam

Nozzle moves, nothing extrudes — clicking as the motor skips backwards.

Heat creep

Heat spreads up the filament path; filament swells and binds before the melt zone.

PLA is fragile here

PLA's low glass point makes it the most heat-creep-prone material.

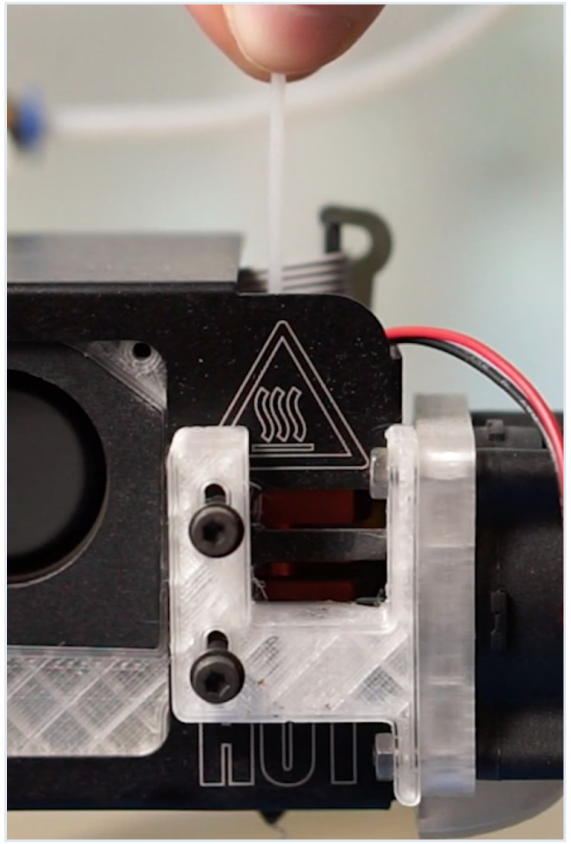
Prevention

Cooling fan on the heatbreak, retraction not overtuned, quality filament.

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A3 · PROCEDURE

Clearing a Clogged Nozzle



Clearing a clog with a needle — hot end, pliers, patience

Hot needle
Heat the nozzle, insert a needle/wire held with pliers — never fingers.

Gentle pressure
Poke the clog while pressing the filament lightly from above.

Cold pull
Once flowing, cool slightly and yank the filament to extract residue.

Persistent clogs
Swap the nozzle — they are consumables, not heirlooms.

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MATERIALS · K2

The Filament Universe

1

Standards

PLA · ABS · PETG/PET · Nylon — the four you must know (next slides).

2

Flexibles

TPE / TPU — rubber-like parts, gaskets, tyres.

3

Composites

Carbon-fibre, wood, sandstone, metal-filled — stiffness or looks.

4

Soluble

PVA (water) and HIPS (limonene) — dissolvable support materials.

5

Functional

Conductive, magnetic iron, glow-in-the-dark, Amphora.

6

Engineering

PC, PEEK, ULTEM — high temperature, high strength, high difficulty.

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ABS vs PLA

ABS

- ABS — engineering workhorse
  - Durable, strong, lightweight and slightly flexible
  - Cheap and proven — the professional's default for years
  - Petroleum-based, not biodegradable
  - High temperatures; warps without heated bed/enclosure
  - Unpleasant fumes — ventilate the room

PLA

- PLA — the friendly starter
  - Plant-derived; sweet smell, no harsh fumes
  - Easy to print, minimal warping — ideal for beginners
  - Huge colour/effect range: glow, translucent, silk
  - Brittle vs ABS; softens in a hot car
  - Attracts moisture — store dry

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PETG vs Nylon

PETG

- PETG — tough all-rounder
  - Strong, flexible, water- and chemical-resistant
  - Low warp, low shrink; food-safe grades exist
  - Doesn't absorb moisture from the air readily
  - Needs fine-tuned nozzle & bed temperatures
  - Strings more than PLA while tuning

Nylon

- Nylon — engineering flexibility
  - Strong, durable and flexible; less brittle than ABS/PLA
  - 100% thermoplastic — re-meltable without bond loss
  - High melting temperature required
  - Emits fumes when overheated — ventilate
  - Extremely hygroscopic — dry-box storage essential

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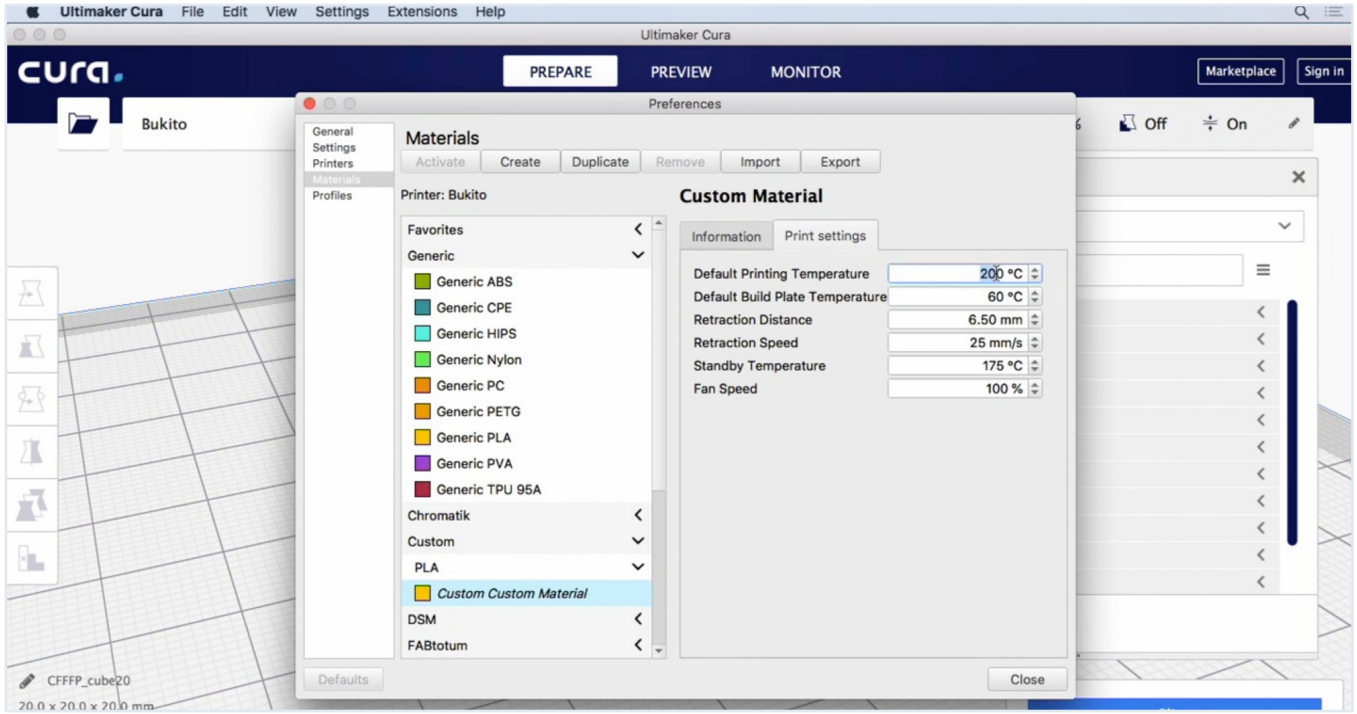
Required Temperatures by Material

Material	Extruder	Bed
PLA	195–215 °C	Room temp or 60 °C
PETG	240–260 °C	Room temp small parts, or 80–90 °C
Nylon	240–260 °C	Room temp, but needs special platforms
ABS	240–260 °C	110–120 °C
Polycarbonate	280–300 °C	130 °C+

Start mid-window; tune in 5 °C steps — strings mean too hot, poor bonding means too cold (Lab 10).

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Material Settings in Cura



Cura material management — the definition your prints inherit

Material definition

Temperature, fan and flow defaults come from the material profile.

Settings override

A value set in Print Settings beats the material definition for that print.

Profiles persist

Create/update a profile to make your tuned overrides permanent.

One source of truth

Keep tuned profiles per material — the end of guesswork.

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Hybrid Metal + Non-Metal Components

1. SPLIT — separate the design into metallic and non-metallic parts
2. SELECT — DMLS/SLM for metals; FDM/SLS for polymers
3. POST-PROCESS — apply each material's finishing chain separately
4. BOND — join with the right technique, e.g. electroless deposition
5. VERIFY — test the assembly's integrity as one component

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WHAT YOU'LL DO · LABS 7–10

Hands-On Labs — Topic 2

Toolchain

- Lab 7: Install Cura
- Slice your first model
- Read the G-code preview

Adhesion & support

- Lab 8: Skirt vs brim vs raft
- Lab 9: Supports for overhangs

Materials

- Lab 10: Filament diagnostics
- Diameter, temperature, flow

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TOPIC 02 · HANDS-ON

Lab 7 — Install Cura and Slice Your First Model

LAB 7

A2 — set up the slicing toolchain used to analyse and fix AM process issues

Install Ultimaker Cura, add your printer profile, and slice a calibration cube to understand how the slicer turns an STL into G-code printer instructions.

You'll build

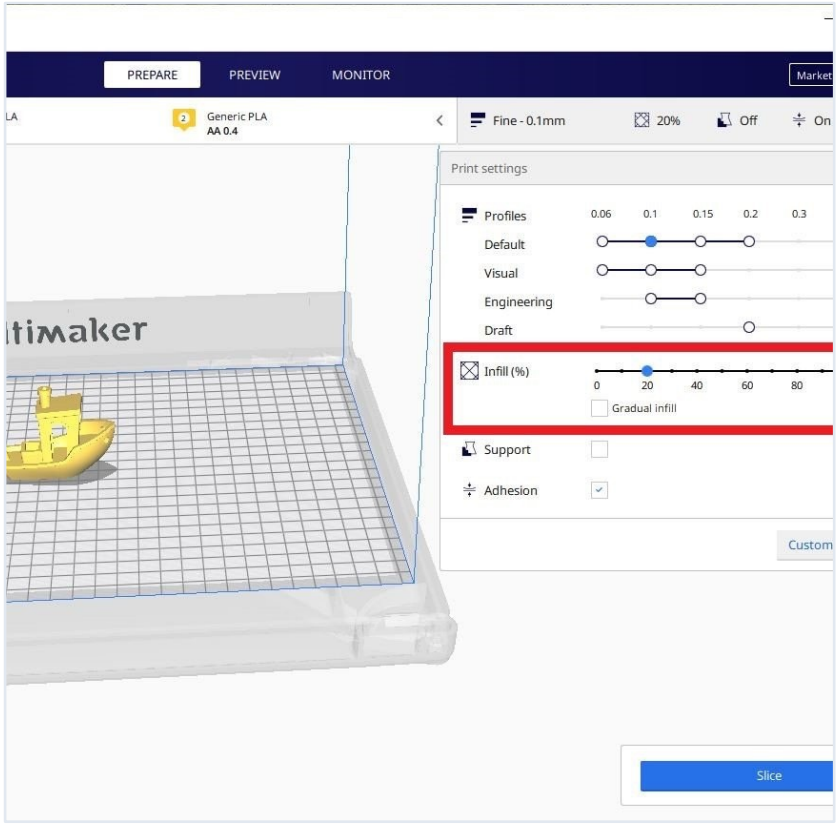
A working Cura installation with your printer profile, and your first sliced G-code file.

Tools: Ultimaker Cura, cube10.stl, cube20.stl

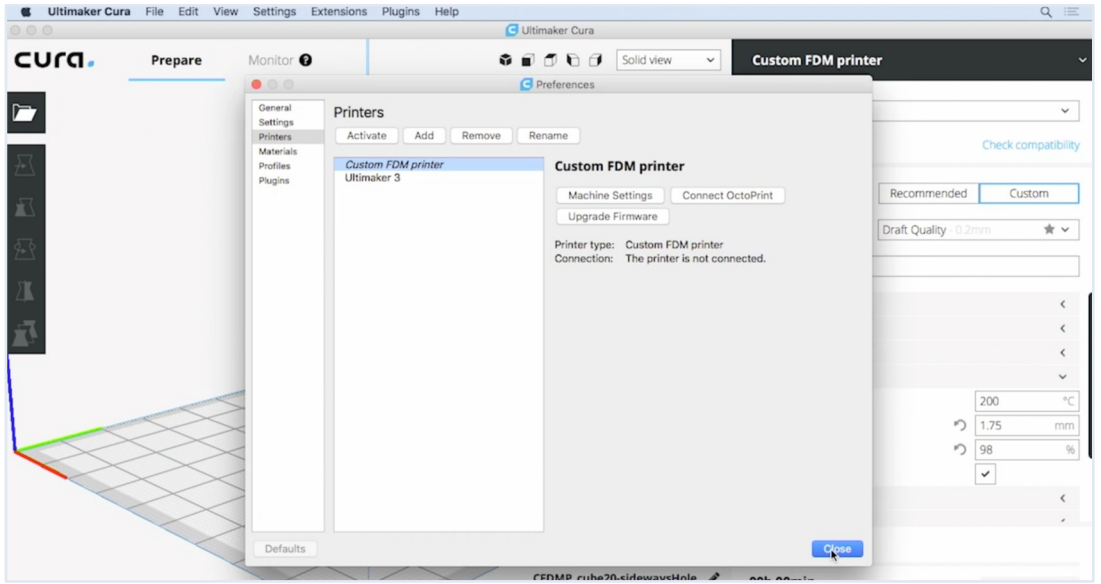
Step-by-step guide: Learner Guide, Lab **7** · labs/lab-07 folder

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Lab 7 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 7 · VERIFY

Lab 7 — Install Cura and Slice Your First Model

✅ Test it

Cura shows per-layer toolpaths for both cubes, cube20.stl measures 20 × 20 × 20 mm, and your comparison explains why doubling linear size increases print time and material by much more than two times.

Full steps and troubleshooting: Learner Guide · Lab **7**

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TOPIC 02 · HANDS-ON

Lab 8 — Bed Adhesion — Skirt vs Brim vs Raft

LAB 8

A2 — analyse first-layer adhesion issues and select the correct adhesion aid

Slice three instances of the same cube with a skirt, a brim and a raft, print them, and compare adhesion strength, removal effort and bottom-surface finish.

You'll build

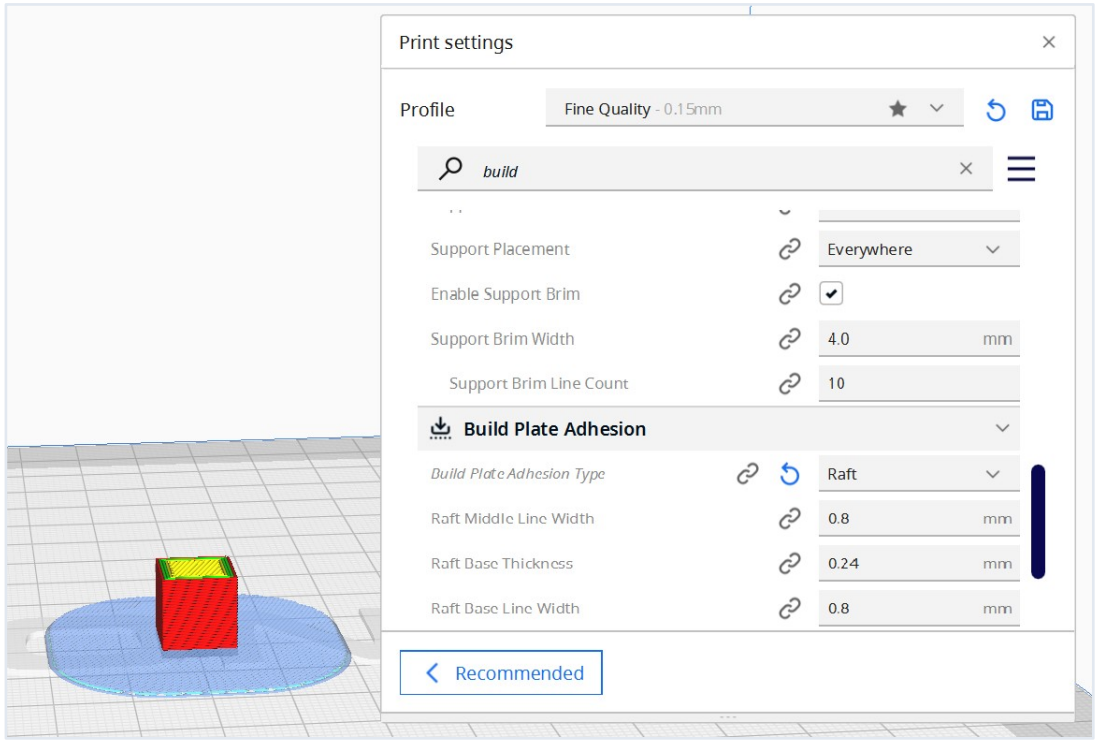
Three prints of cube10.stl demonstrating skirt, brim and raft — and a rule for when to use each.

Tools: Ultimaker Cura, cube10.stl, FDM printer

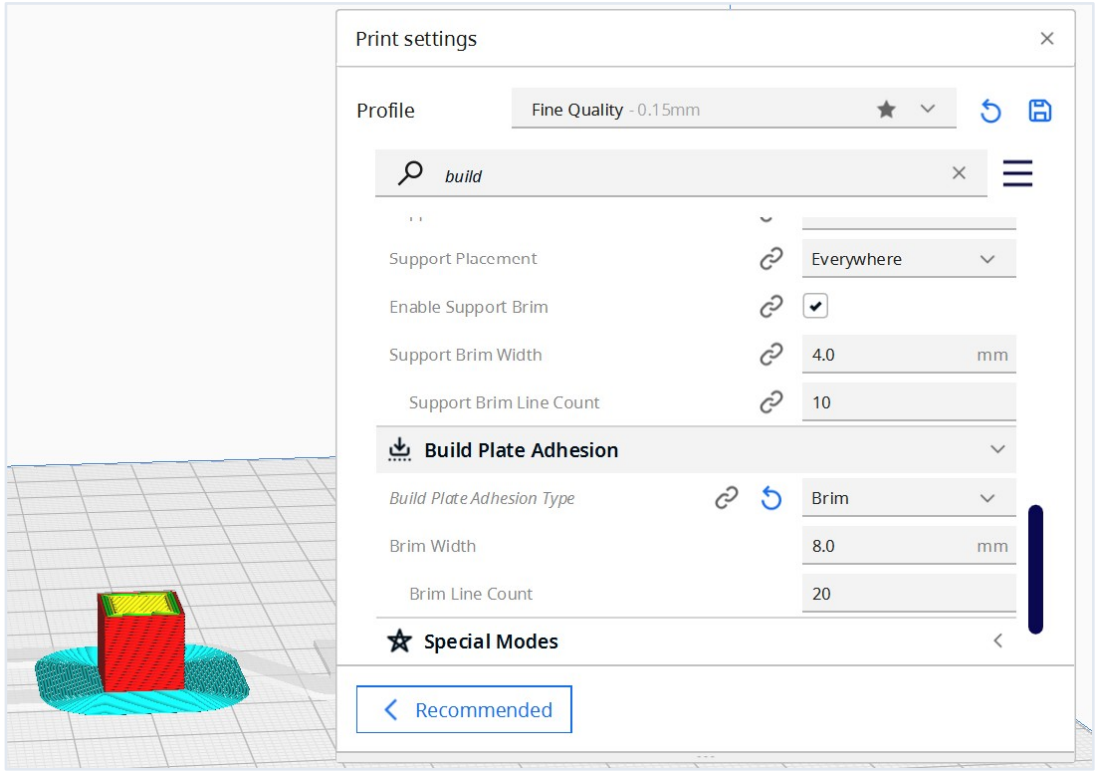
Step-by-step guide: Learner Guide, Lab **8** · labs/lab-08 folder

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Lab 8 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 8 · VERIFY

Lab 8 — Bed Adhesion — Skirt vs Brim vs Raft

✅ Test it

You can state when each aid applies — skirt to prime the nozzle, brim for warping-prone or small-footprint parts, raft for top-heavy or poorly adhering parts — and show the bottom-finish difference.

Full steps and troubleshooting: Learner Guide · Lab **8**

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TOPIC 02 · HANDS-ON

Lab 9 — Support Structures for Overhangs

LAB 9

A2, A3 — determine when overhang geometry requires supports and generate them

Use supplied twist, wedge and sideways-hole models to identify unsupported islands, compare orientation choices, then generate only the supports the geometry needs.

You'll build

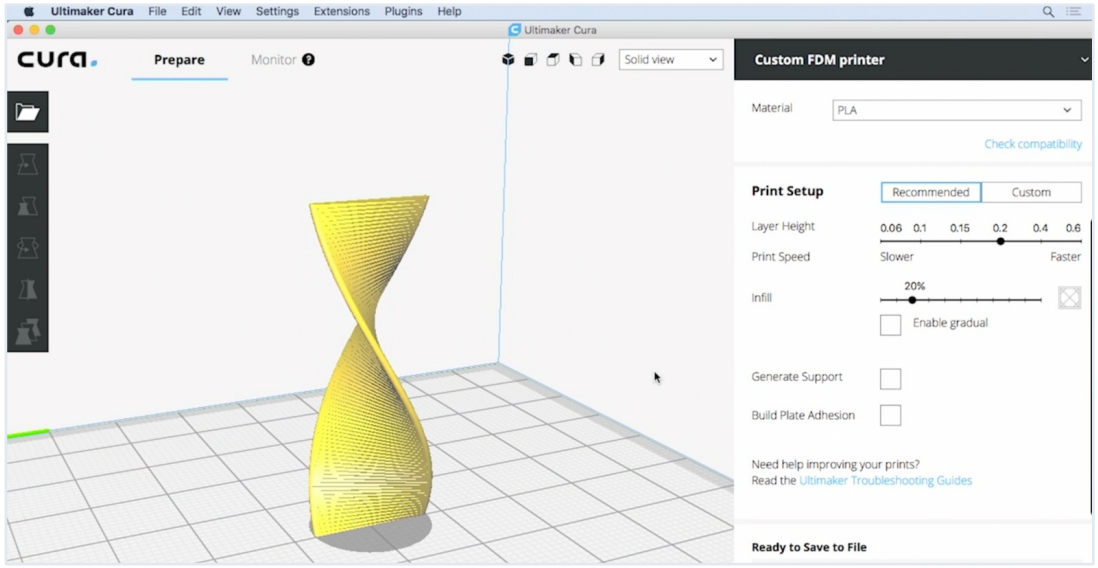
A support decision sheet plus a printed Twisted Solid whose critical overhangs succeed with controlled support use.

Tools: Ultimaker Cura, Twisted-Solid.stl, wedge.stl, cube20-sideways-hole.stl, FDM printer

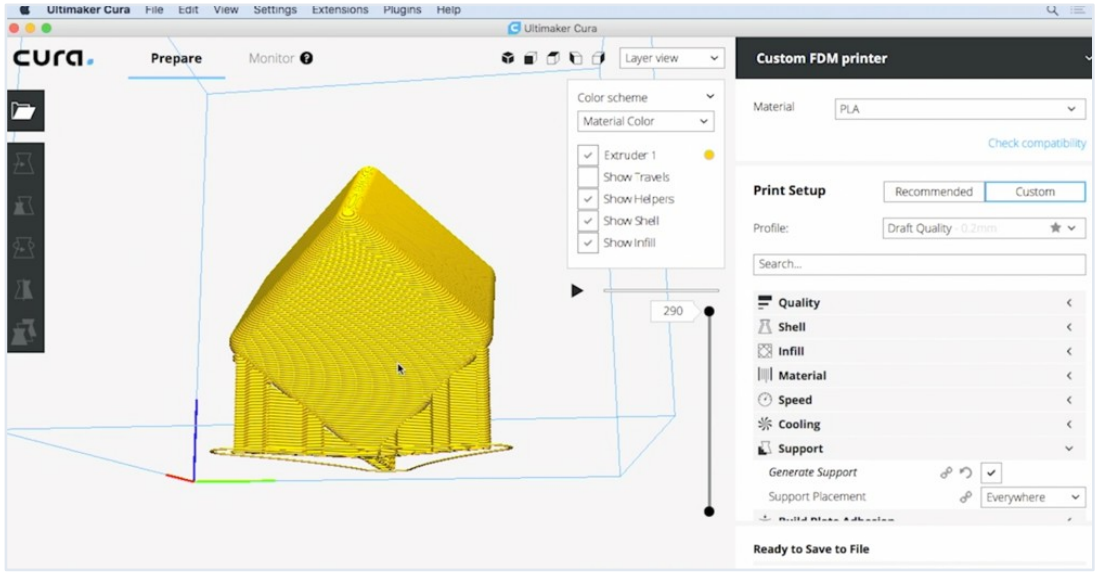
Step-by-step guide: Learner Guide, Lab **9** · labs/lab-09 folder

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Lab 9 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 9 · VERIFY

Lab 9 — Support Structures for Overhangs

✅ Test it

Your decision table correctly identifies unsupported islands and distinguishes when support is necessary from when reorientation is the better fix; the Twisted Solid print completes with removable support and only minor underside scarring.

Full steps and troubleshooting: Learner Guide · Lab **9**

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TOPIC 02 · HANDS-ON

Lab 10 — Filament Diagnostics — Diameter, Temperature and Flow

LAB 10

A2, A3 — analyse material-side causes of extrusion issues and verify filament quality

Measure filament diameter with digital calipers, verify it against the slicer's material setting, and run a temperature check to place your filament in its working window.

You'll build

A verified filament profile plus retraction and cooling evidence from purpose-built diagnostic models.

Tools: Digital calipers, filament spool (PLA/ABS/PETG), Ultimaker Cura, retraction-test.stl, cooling-spire.stl, FDM printer

Step-by-step guide: Learner Guide, Lab **10** · labs/lab-10 folder

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TOPIC 02 · HANDS-ON

Lab 10 — What You'll Work With



Material	Extruder	Bed
PLA	195–215°C	Room temp or 60°C
PETG	240–260°C	Room temp for small objects, or 80–90°C
Nylon	240–260°C	Room temp, but requires special platforms
ABS	240–260°C	110–120°C
Polycarbonate	280–300°C	130°C+

Reference — the model / setting you'll produce

Reference — as prepared in the slicer / tool

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LAB 10 · VERIFY

Lab 10 — Filament Diagnostics — Diameter, Temperature and Flow

✅ Test it

Your diameter average is within ±0.05 mm of nominal, Cura carries the measured value, and the diagnostic models support a retraction/cooling choice with less stringing and a cleaner spire without introducing gaps or jams.

Full steps and troubleshooting: Learner Guide · Lab **10**

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Recap — Topic 2

- AM issues trace to five sources: hardware, materials, process conditions, environment and design.
- The slicer converts STL to G-code — adhesion aids (skirt/brim/raft) and supports are slicing decisions.
- Extrusion faults, retraction, oozing, jams and heat creep are diagnosed from the printed evidence.
- Each material demands its temperature window, bed and storage — verified, not assumed (Lab 10).
- Hybrid metal/polymer components pair the right AM process per part with the right bonding technique.

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03

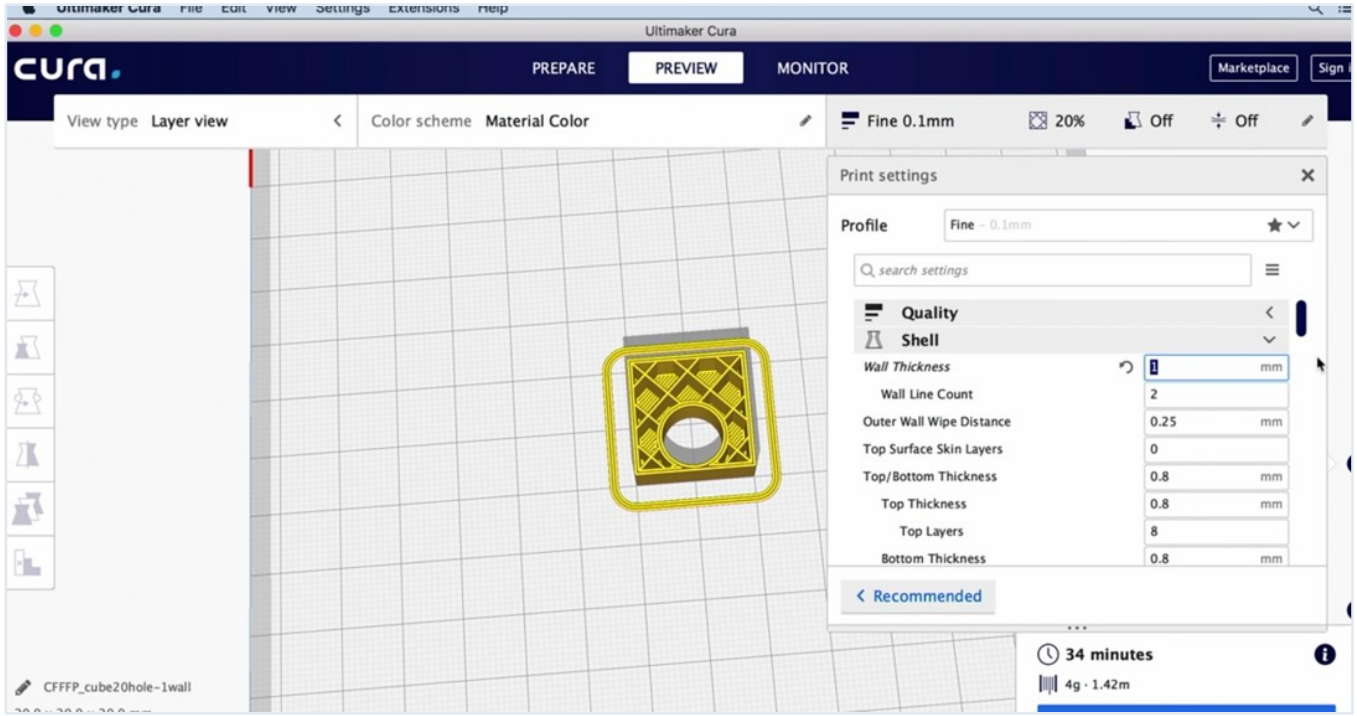
TOPIC 03

Process Improvement for AM

Infill · surface quality · supports · post-processing · business impact (K5, K7 · A5, A6)

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Infill and Perimeters



Cura print settings — walls and infill density

Perimeter (walls)
The solid outline around each layer — walls carry most of a part's strength.

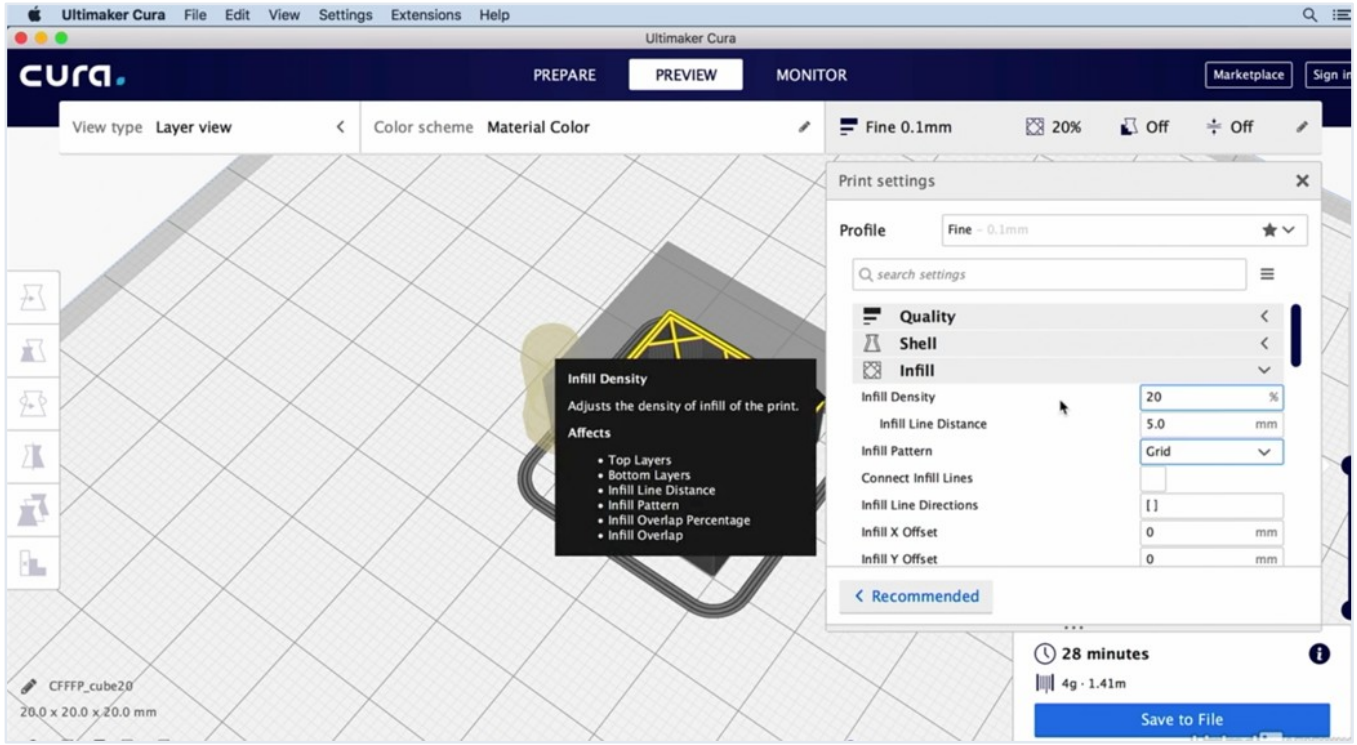
Infill
The internal lattice — typically only 15–20% of the enclosed volume.

The dial
More walls/infill: stronger, heavier, slower. Less: faster, lighter, weaker.

Lab 11
You'll measure exactly what each setting costs in time and grams.

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Infill — Density and Pattern Effects



Infill preview at increasing densities

12–20% typical

Fraction of the interior that is plastic — the default band for most parts.

50%+ for strength

Functional, loaded parts justify denser infill and less warping.

Pattern matters

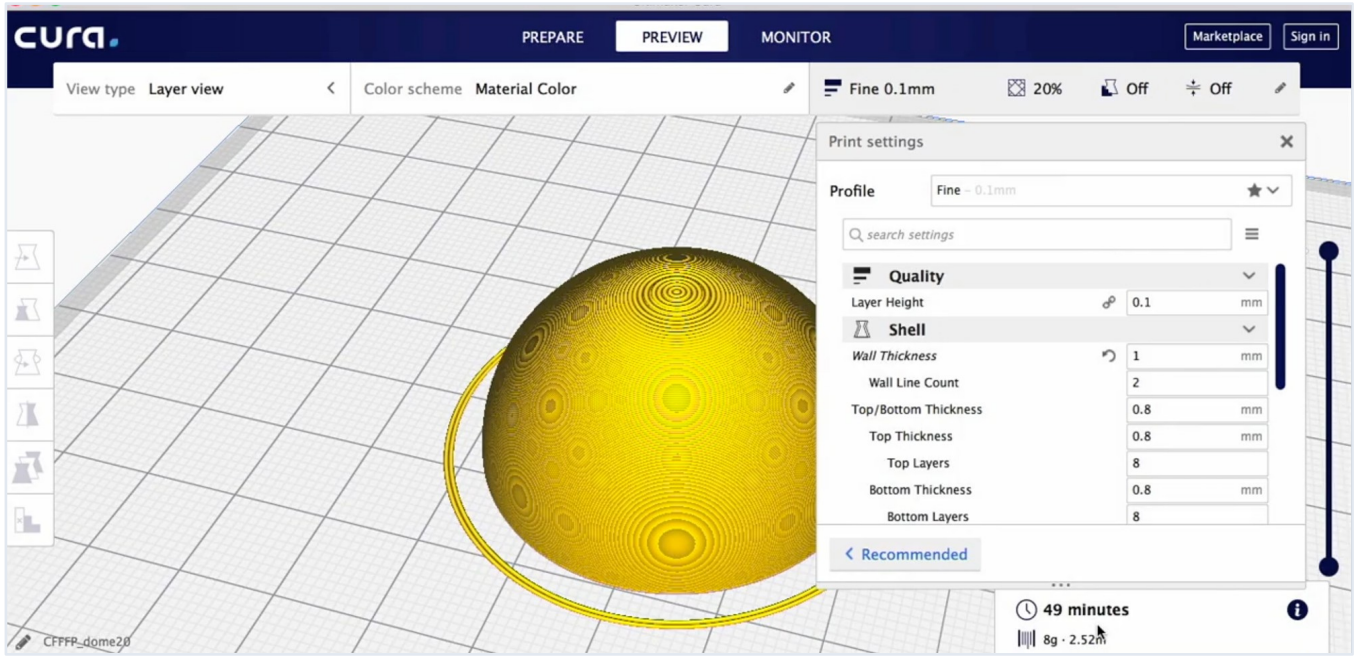
Grid, zigzag, gyroid, cubic — each trades strength direction vs speed.

Lab 12

Zigzag 20% vs grid 30% — printed and squeezed side by side.

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Surface Quality — Layer Height



Layer-height setting drives finish and duration

The resolution dial

Layer height sets the visible stair-step on every curved or sloped face.

Time trade

0.1 mm looks superb and takes twice as long as 0.2 mm; 0.3 mm is draft mode.

Top thickness follows

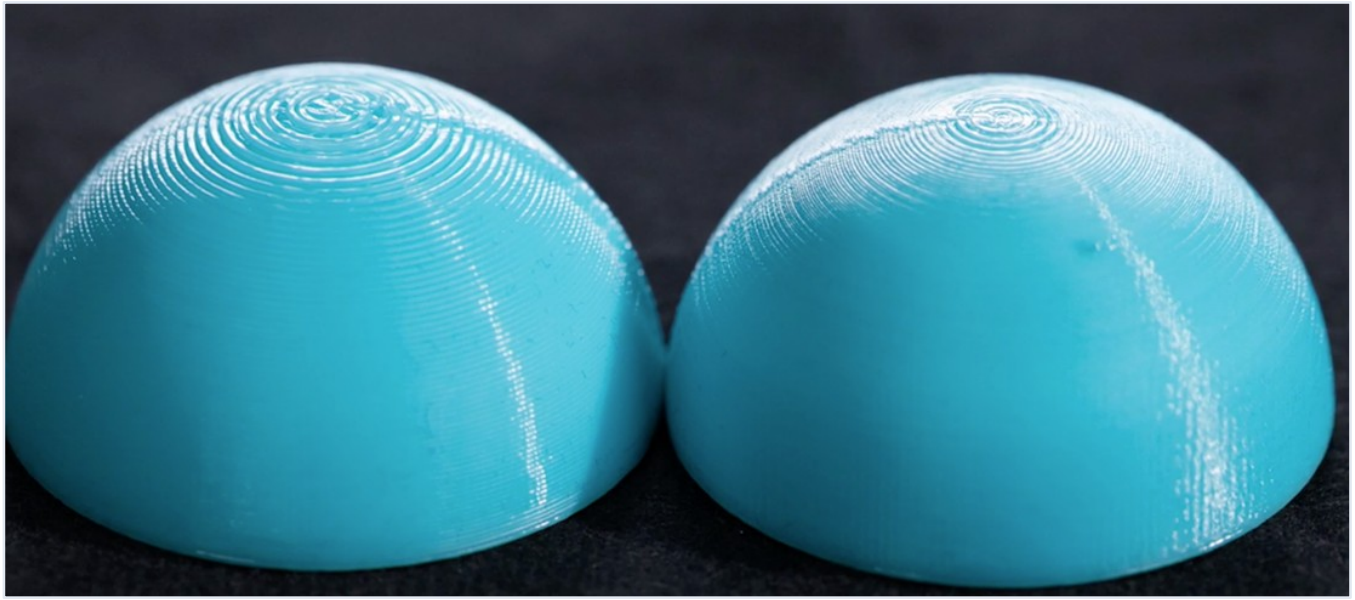
Keep top THICKNESS constant so the layer count adapts — Lab 13.

Match to purpose

Draft iterations coarse; presentation parts fine.

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The Top Layer — Pillowing



Pillowing: the roof sags into sparse infill

Preview lies
The simulated top looks flat; physics prints it over air gaps.

Pillowing
Thin roofs sag between infill lines, or curl and break — bumpy tops.

Fixes
More top layers, denser infill under roofs, better cooling.

Rule of thumb
Top thickness ≥ 4× layer height for a clean roof.

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A5 · FINISH

Surface Texture Options



Surface finish elevates a print into a product

Build surface finish

Glossy or textured bottom faces — decided by the bed surface itself.

Ironing — Lab 14

The hot nozzle re-passes top faces at ~10% flow for glass-smooth roofs.

Fuzzy skin — Lab 15

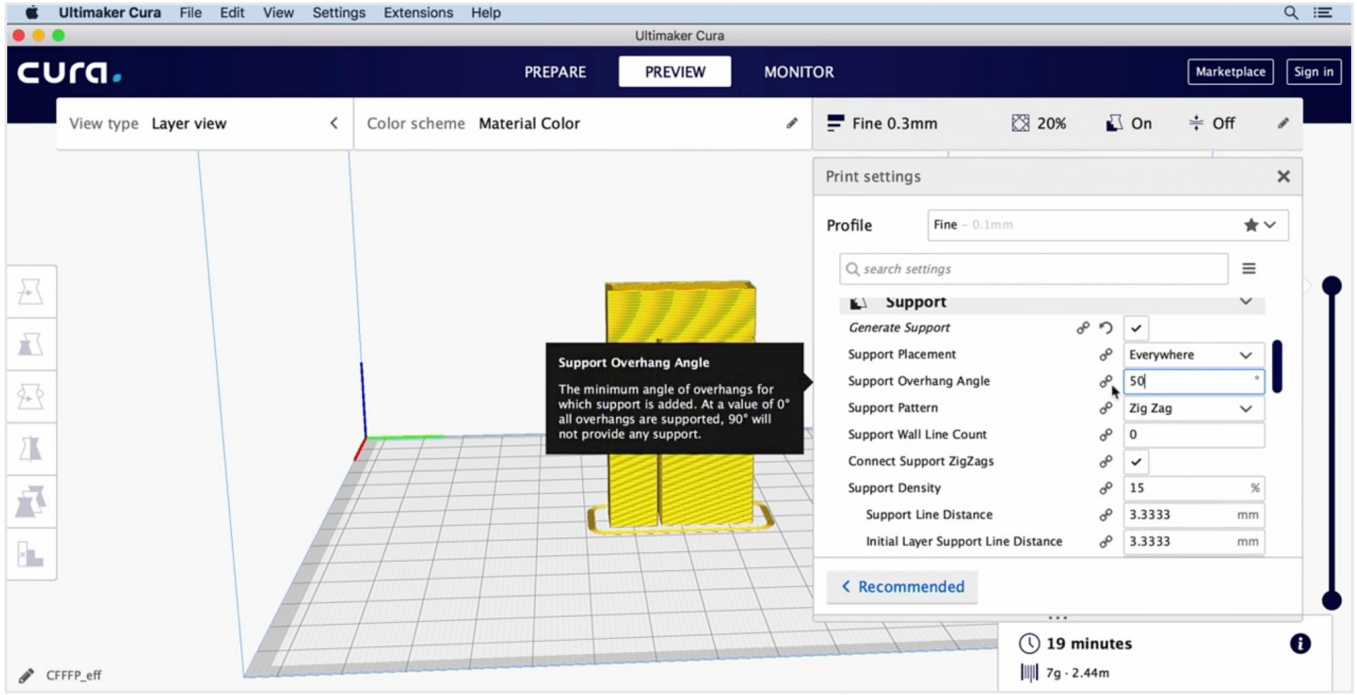
Random jitter on outer walls hides layer lines behind texture.

Deliberate design

Texture is a slicing decision, not an accident.

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Support Strategy



Cura support generation — angle and placement settings

Overhangs need support

Anything printing into free space beyond ~45–50° needs scaffolding.

Reorient first

Rotating the part is free — supports cost material, time and finish.

Placement control

Touching-buildplate vs everywhere; blockers protect chosen faces.

Removal is a cost

Every support leaves a scar — plan removal before slicing (Lab 16).

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A5 · JUDGEMENT

Support Challenges



A part whose internal void must stay support-free

Inaccessible cavities
Supports inside holes and channels may be impossible to extract.

Skip or limit
Sometimes the right call is limiting supports and accepting minor sag.

Design around it
Chamfer the overhang, bridge the gap, or split the part for printing.

The craft
Support judgement separates operators from process engineers.

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Pros and Cons — SLS & Inkjet-Style AM

Powder + laser

- SLS
  - No supports — powder holds the part
  - Tough, functional nylon parts; batch nesting
  - Grainy surface; powder handling & recycling
  - Long cool-down before parts emerge

Droplets + binder

- Material/Binder jetting
  - Multi-material and full-colour capability
  - Fine features, smooth surfaces
  - Fragile green parts (binder); resin cost
  - Nozzle clogging and layer misalignment risks

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Pros and Cons — DMLS & SLM

Sinter

- DMLS — sintered metal
  - Complex metal geometries, good strength
  - Established in dental & aerospace
  - Warping from rapid cooling; powder inconsistency
  - Expensive machines and powders

Melt

- SLM — fully melted metal
  - Full density, best mechanical properties
  - Single-alloy purity
  - Oxidation risk — inert atmosphere required
  - High residual stresses — heat treatment mandatory

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Business Considerations for AM Operations

1. **Production speed**
   Compare cycle times before and after every process change.

2. **Quality**
   Functionality, mechanical properties, dimensional accuracy, surface finish.

3. **Material usage**
   Consumption and wastage — supports, purge, failed prints.

4. **Resource utilisation**
   Machine uptime, operator hours, floor space, energy.

5. **Cost & savings**
   Full cost delta of the change, capital to consumables.

6. **Ease of use & feedback**
   Operator experience and the challenges users report.

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K5 · THE LOOP

Assessing a Process Improvement

1. BASELINE — record speed, quality, material and cost before
2. CHANGE — introduce one improvement at a time
3. MEASURE — same metrics, same parts, after
4. COMPARE — quantify the delta; collect operator feedback
5. DECIDE — adopt, tune or roll back; document the standard

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Sustaining AM Operations Long-Term

1

Calibration & maintenance

Regular calibration and preventive maintenance of AM equipment.

2

People

Continuous training and education of AM personnel.

3

Materials R&D

Ongoing research into sustainable, environmentally-friendly materials.

4

Collaboration

Work with stakeholders on sustainability standards and best practices.

5

Circularity

Recycle powders and failed prints; design for material recovery.

6

Knowledge base

Document tuned profiles and fixes — institutional memory compounds.

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Post-Processing — FDM

1

Support removal

Manual removal with tools, or dissolve PVA in water / HIPS in limonene.

2

Filling & sanding

Epoxy fillers close gaps; progressive grits smooth surfaces (Lab 16).

3

Vapour smoothing & dipping

Solvent vapour or acetone dip melts the surface to a gloss.

4

Priming & painting

Primer then paint for presentation parts.

5

Coating

Epoxy resin coats add a protective, sealed skin.

6

Metal plating

Chemical or electro-plating gives polymer parts a metallic finish.

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Post-Processing — SLS and SLA

1

SLS: cooling

Hours of controlled cool-down before parts leave the build chamber.

2

SLS: clean-up

Compressed air, vacuum, sand blasting or brushing recover unsintered powder.

3

SLS: finishing

Sanding, polishing, dyeing, painting, coatings, electroplating, vapour smoothing.

4

SLA: wash & cure

IPA wash removes resin; UV post-cure completes polymerisation.

5

SLA: finishing

Sanding, polishing, dyeing, painting, coating, electroplating.

6

Safety

Gloves and ventilation — powders and uncured resin are hazardous.

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Post-Processing — Metal AM

1

Sieving & depowdering

Unmelted powder recovered by sieving; trapped powder shaken out by vibration.

2

Heat treatment

Vacuum/inert furnace relieves thermal stress — before build-plate separation.

3

Part cut-off

Chiselling, bandsaw or wire EDM separate the part from the plate.

4

Support removal

Chiselling, wire EDM, CNC machining or thermal deburring by geometry.

5

Machining

Threads, holes and precision flats machined to tolerance.

6

Surface finish

Shot blasting, plating, polishing, micromachining — per the specification.

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WHAT YOU'LL DO · LABS 11–16

Hands-On Labs — Topic 3

Strength & speed

- Lab 11: Perimeters & infill
- Lab 12: Infill patterns strength test

Surface quality

- Lab 13: Layer height ladder
- Lab 14: Ironing
- Lab 15: Fuzzy skin

Finishing

- Lab 16: Support removal
- Progressive sanding to smooth

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TOPIC 03 · HANDS-ON

Lab 11 — Perimeters and Infill — Strength vs Speed

LAB 11

A5, A6 — apply wall/infill optimisation and assess its measured impact on time and material

Vary wall count and infill density on the same model in Cura, and record how each change moves print time, material use and expected strength.

You'll build

A comparison table of wall/infill settings against sliced print time and material grams.

Tools: Ultimaker Cura, cube10.stl, line-width-test.stl, cube20 hole-wall variants

Step-by-step guide: Learner Guide, Lab **11** · labs/lab-11 folder

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LAB 11 · VERIFY

Lab 11 — Perimeters and Infill — Strength vs Speed

✅ Test it

Your evidence separates wall-count, infill and line-width effects; the matched hole models show how wall planning changes continuity and dimensional accuracy, and you can justify settings for functional versus decorative parts.

Full steps and troubleshooting: Learner Guide · Lab **11**

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TOPIC 03 · HANDS-ON

Lab 12 — Infill Patterns — Zigzag vs Grid Strength Test

LAB 12

A5 — evaluate infill pattern and density as a quality-improvement technique

Print cube10.stl with zigzag 20% and grid 30% infill, tune the top layers, then compare the two cubes' stiffness by hand.

You'll build

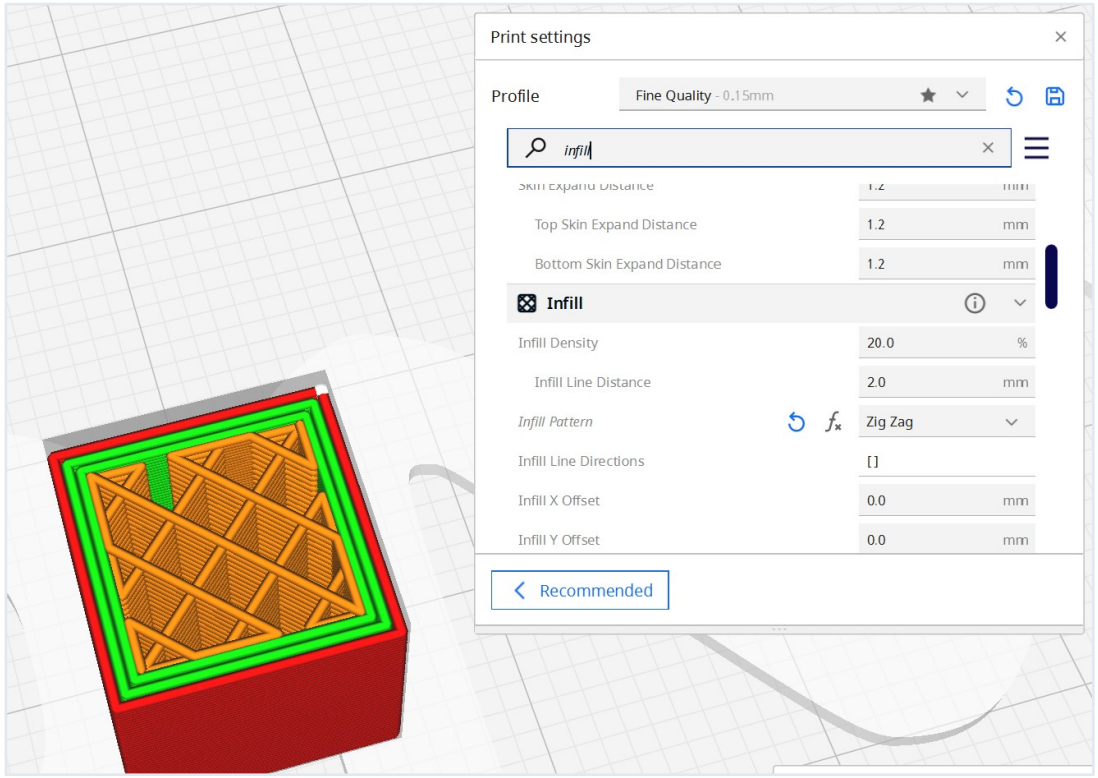
Two printed cubes with different infill patterns/densities and a hands-on strength comparison.

Tools: Ultimaker Cura, cube10.stl, Optimize1–4 model set, FDM printer

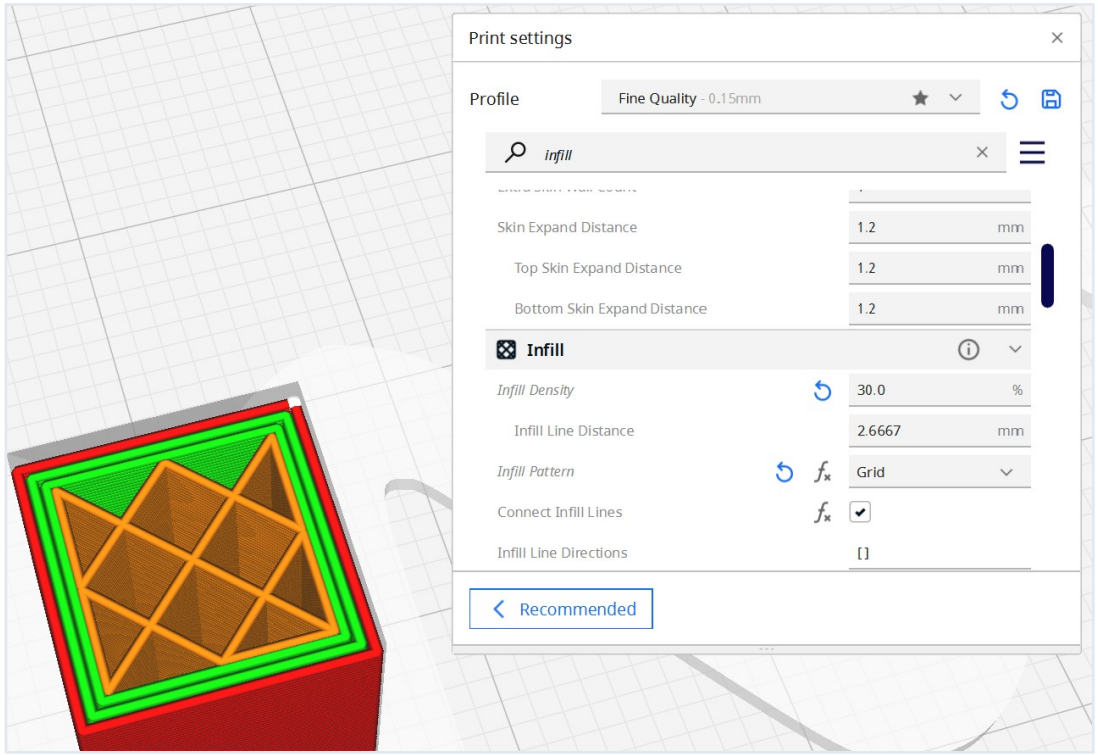
Step-by-step guide: Learner Guide, Lab **12** · labs/lab-12 folder

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Lab 12 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 12 · VERIFY

Lab 12 — Infill Patterns — Zigzag vs Grid Strength Test

✅ Test it

The grid 30% cube is stiffer and better supported, while your four-variant scorecard shows that geometry and orientation can outweigh infill choice; your final recommendation is backed by time, material and support evidence.

Full steps and troubleshooting: Learner Guide · Lab **12**

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TOPIC 03 · HANDS-ON

Lab 13 — Layer Height and Top Thickness — Surface Quality

LAB 13

A5 — optimise layer height and top thickness for surface quality vs speed

Slice the same model at 0.1 mm, 0.2 mm and 0.3 mm layer heights, adjust top thickness to a constant, and compare surface quality against print time.

You'll build

Sliced (and optionally printed) evidence of the layer-height ladder: finish vs time.

Tools: Ultimaker Cura, cube10.stl, sloped-cube.stl, FDM printer

Step-by-step guide: Learner Guide, Lab **13** · labs/lab-13 folder

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LAB 13 · VERIFY

Lab 13 — Layer Height and Top Thickness — Surface Quality

✅ Test it

You can rank the layer heights by finish and time, point to visible stair-stepping on sloped-cube.stl, and explain why constant top thickness—not a fixed top-layer count—keeps the comparison fair.

Full steps and troubleshooting: Learner Guide · Lab **13**

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TOPIC 03 · HANDS-ON

Lab 14 — Ironing — Glass-Smooth Top Surfaces

LAB 14

A5 — apply ironing to improve top-surface quality and assess its cost

Enable Cura's ironing feature, which passes the hot nozzle over the top surface with a tiny extrusion, and compare the top finish with and without it.

You'll build

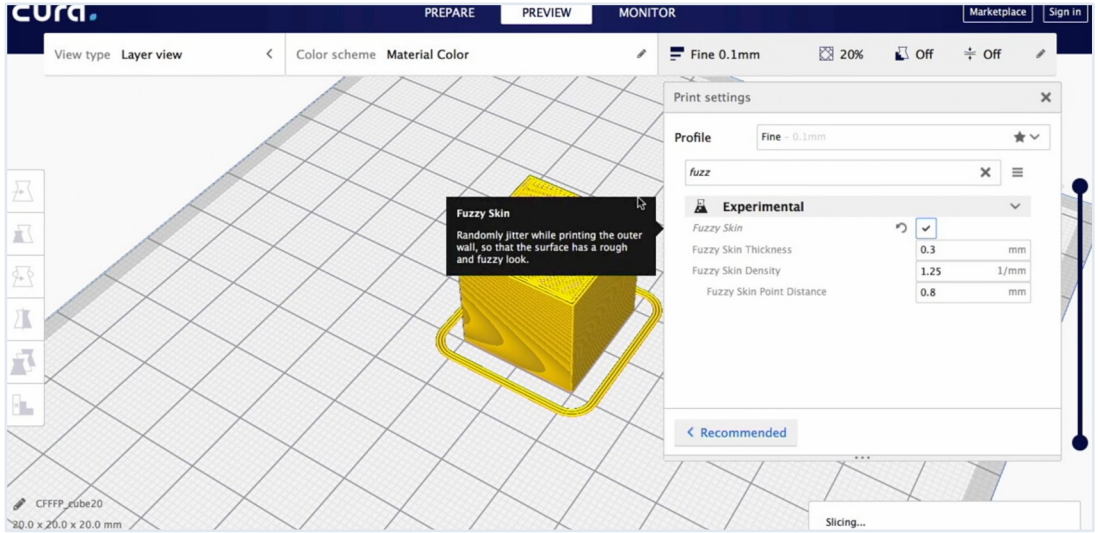
Two cube prints — ironed and standard — demonstrating the top-surface improvement.

Tools: Ultimaker Cura, cube10.stl, z-block.stl, FDM printer

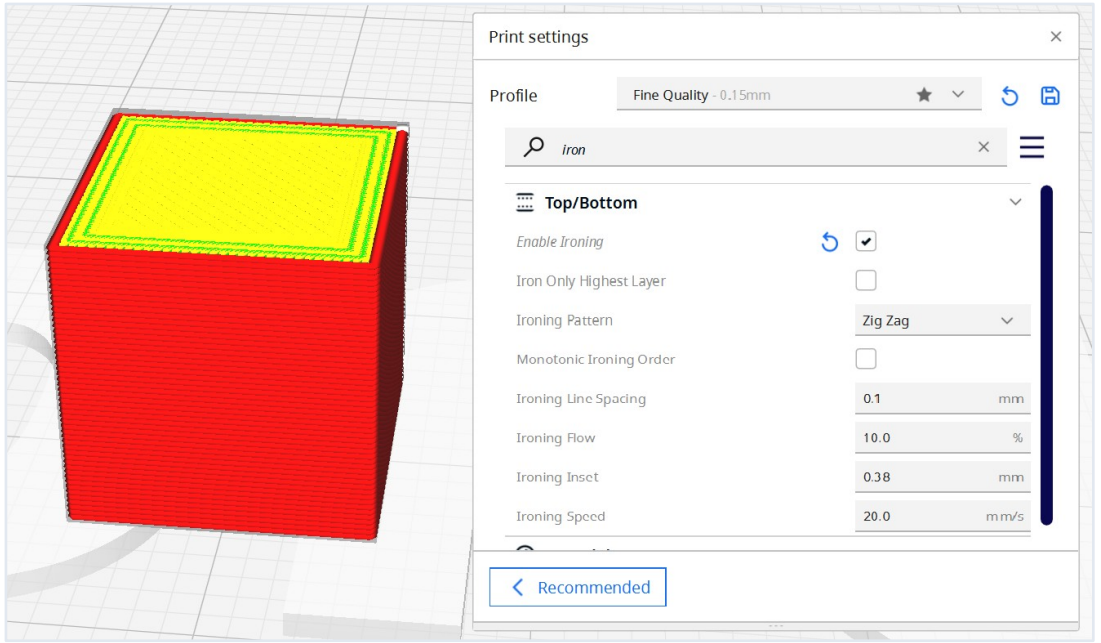
Step-by-step guide: Learner Guide, Lab **14** · labs/lab-14 folder

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Lab 14 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 14 · VERIFY

Lab 14 — Ironing — Glass-Smooth Top Surfaces

✅ Test it

The ironed cube is visibly smoother than the control, z-block.stl shows ironing constrained to eligible top faces, and you can justify the added time for lids, logos and presentation surfaces.

Full steps and troubleshooting: Learner Guide · Lab **14**

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TOPIC 03 · HANDS-ON

Lab 15 — Fuzzy Skin — Textured Walls on a Vase

LAB 15

A5 — apply fuzzy skin as a deliberate surface-texture treatment

Enable fuzzy skin on a vase model so the outer wall is jittered into a stone-like texture that hides layer lines, and print it with buildplate adhesion.

You'll build

A vase with a uniform fuzzy texture that masks layer lines completely.

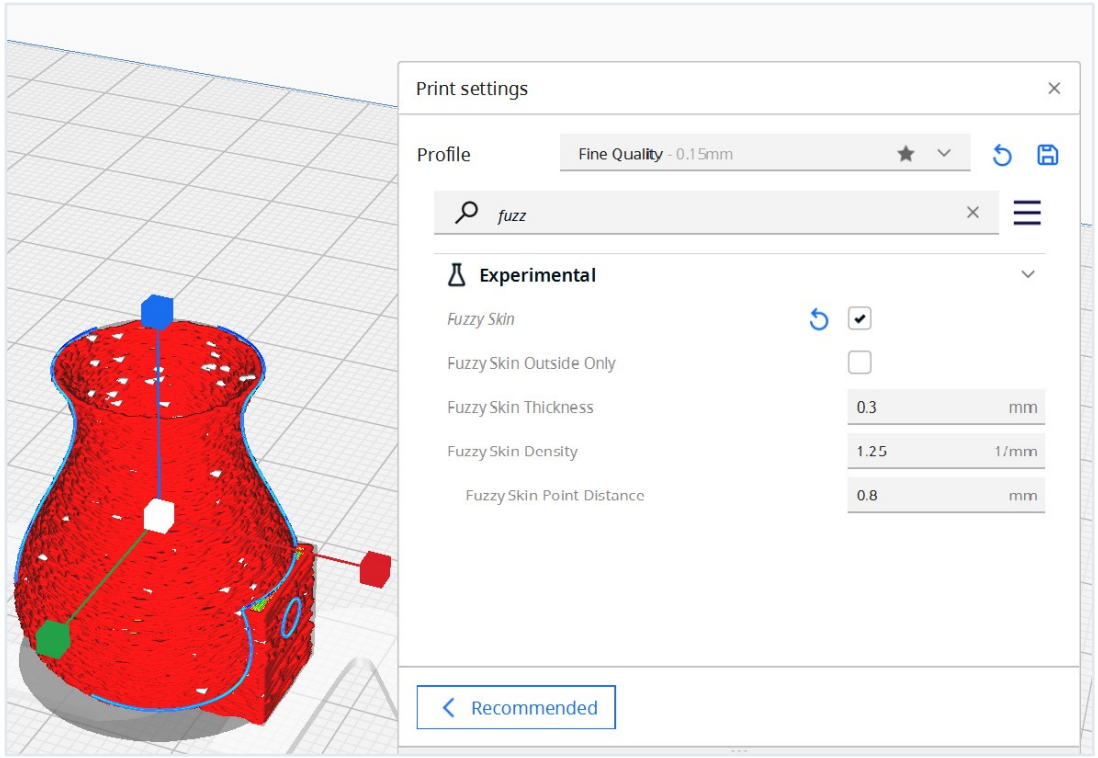
Tools: Ultimaker Cura, open vase.stl, 1.5mm-wall-vase.stl, FDM printer

Step-by-step guide: Learner Guide, Lab **15** · labs/lab-15 folder

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TOPIC 03 · HANDS-ON

Lab 15 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 15 · VERIFY

Lab 15 — Fuzzy Skin — Textured Walls on a Vase

✅ Test it

The printed vase carries an even tactile texture, and the 1.5 mm diagnostic vase retains continuous walls in Preview—showing that the texture setting fits the available wall thickness.

Full steps and troubleshooting: Learner Guide · Lab **15**

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TOPIC 03 · HANDS-ON

Lab 16 — Post-Processing — Support Removal and Sanding

LAB 16

A5 — apply post-processing to lift the quality of an FDM part

Print the vase tilted so it needs supports, remove them cleanly, then sand the scarred surfaces through progressive grits to a smooth finish.

You'll build

A tilted-print vase taken from raw support-scarred print to a smooth finished part.

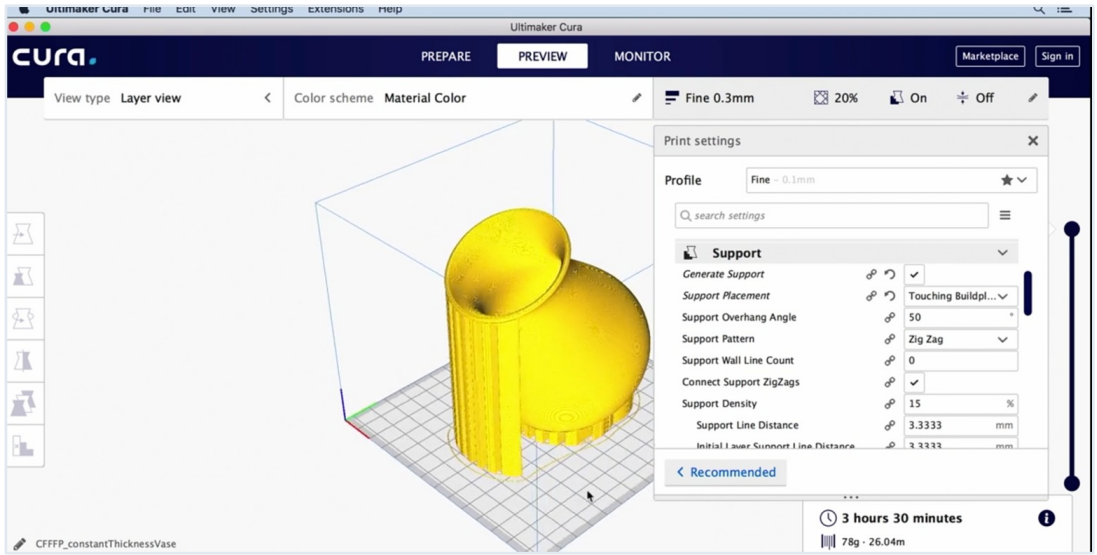
Tools: Ultimaker Cura, open vase.stl, test-part.stl, FDM printer, pliers, sandpaper 120→400 grit

Step-by-step guide: Learner Guide, Lab **16** · labs/lab-16 folder

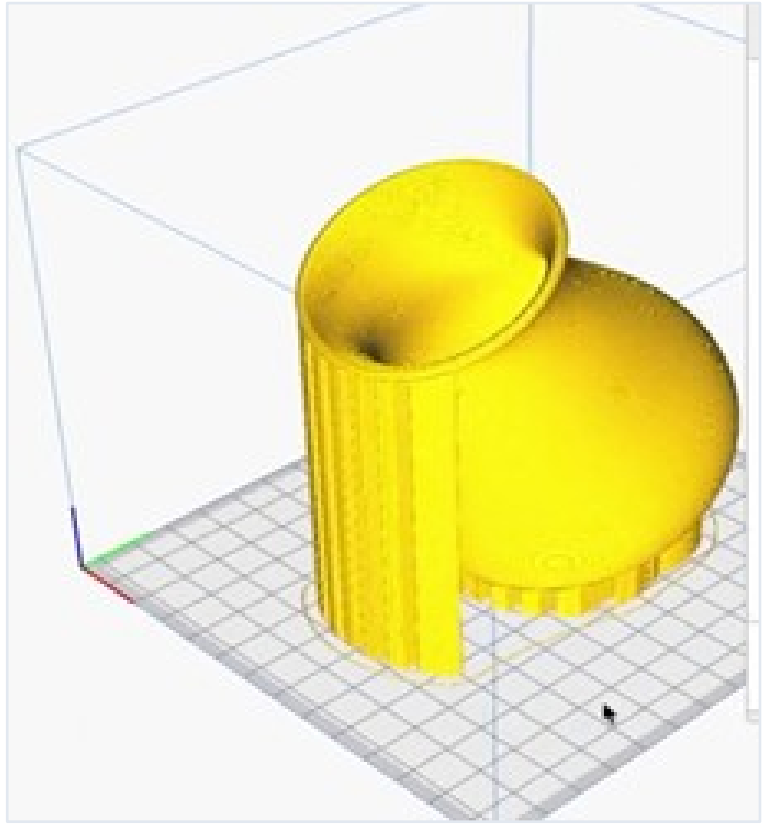
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TOPIC 03 · HANDS-ON

Lab 16 — What You'll Work With



Reference — the model / setting you'll produce



Reference — as prepared in the slicer / tool

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LAB 16 · VERIFY

Lab 16 — Post-Processing — Support Removal and Sanding

✅ Test it

The support side is smooth from a normal viewing distance, and your evidence shows a planned 120→240→400 grit process that removes scars without rounding protected features or critical dimensions.

Full steps and troubleshooting: Learner Guide · Lab **16**

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Recap — Topic 3

- Walls and infill are the strength-vs-speed dial; measure what every setting costs before standardising it.
- Layer height, top thickness, ironing and fuzzy skin put surface quality under deliberate control.
- Reorient first, support second — and plan support removal before slicing, not after printing.
- Assess every improvement with the baseline→change→measure→compare→decide loop across speed, quality, material, cost and feedback.
- Calibration, training, sustainable materials and collaboration keep an AM operation viable long-term.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES DELIVERED

What You Achieved

1

Analysed AM systems

Seven process families, materials, benefits and trade-offs — and when to choose which.

2

Designed for AM

Extrude, sweep, loft, revolve, tolerances and stress-aware design in Fusion 360.

3

Reverse engineered

3D-scanned a real object and rebuilt scans into printable CAD.

4

Diagnosed process issues

Adhesion, extrusion, temperature, jams and materials — evidence-based fixes.

5

Optimised processes

Infill, surface quality, supports and post-processing — measured, not guessed.

6

Assessed business impact

Speed, quality, material, cost and feedback before and after each change.

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SUPPLIED STL / FUSION / CURA ASSETS · LABS 5 & 7–16

Diagnostic Model Library

1

Fit & tolerance

FDM Box Fusion/Cura projects and STL — inspect bodies, orientation, clearance and slicer intent.

2

Support & orientation

Twisted Solid, wedge and sideways-hole cube — distinguish unsupported islands from geometry solved by rotation.

3

Flow & cooling

Retraction towers, cooling spire and combined cooling tower — change one setting and read the printed evidence.

4

Walls & holes

Line-width strip and matched hole-wall cubes — test continuity, wall planning and dimensional accuracy.

5

Optimisation

Four supplied variants — compare support, time and material before selecting the best printable design.

6

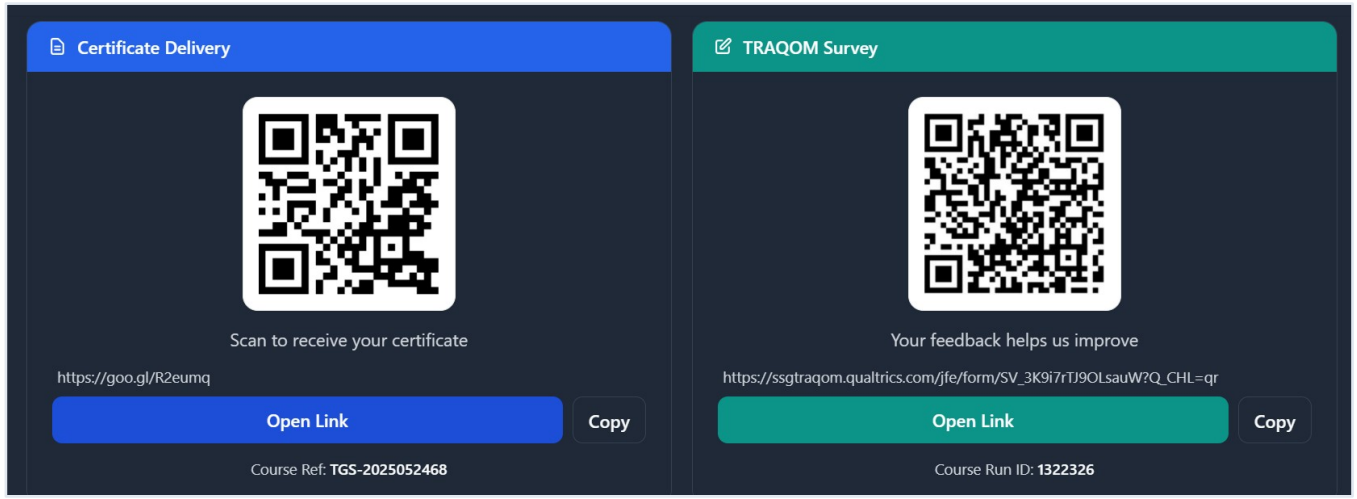
Finish & features

Sloped cube, Z block, thin-wall vase and mixed test part — expose stair-stepping, ironing, textures and scars.

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WRAP-UP · MANDATORY

Certificate & TRAQOM Survey



Certificate delivery and TRAQOM survey QR codes on the LMS

Certificate of Achievement

Issued on 75% attendance + Competent assessment — download from the LMS.

OpenCert

SkillsFuture Singapore issues an OpenCert Statement of Attainment.

TRAQOM survey

Scan the survey QR — your feedback is mandatory for WSQ funding.

Portal

https://lms-tms.tertiaryinfotech.com

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KEEP BUILDING

Recommended Next Courses

- WSQ — Product Design with Fusion 360
- WSQ — Product Design with Autodesk Inventor
- WSQ — AutoCAD Civil 3D for Infrastructure Design
- WSQ — Basic Technical Drawing with SketchUp for Beginners
- WSQ — 3D Modelling with Blender for Beginners

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WE'RE HERE TO HELP

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: 6100 0613
- Web: www.tertiarycourses.com.sg

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FINAL ASSESSMENT

Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) — 1 hour, open book.

2

Practical Performance (PP)

Scenario-based practical tasks — 90 minutes, open book.

3

Open book

Slides, Learner Guide and approved materials only.

4

Digital attendance

Remember to take the Assessment digital attendance (TRAQOM).

5

Eligibility

Minimum 75% attendance to be eligible for assessment and funding.

6

Upload

Submit your completed answers on https://lms-tms.tertiaryinfotech.com.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then the Practical Performance — open book
4. Upload your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

1

AM · PM · Assessment

It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.

2

SSG QR code

The trainer/administrator displays the digital attendance QR code generated from the SSG portal.

3

Scan & submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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HAPPY PRINTING

Thank You!

You are now ready to innovate product development with 3D printing and additive manufacturing.

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